MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE CHF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

* 1.2 Classification This specification covers one class of modules

Class 1 as specified in MIL-STD-1389

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules

MIL-STD-1665 Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.936 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CHF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/71-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4 QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

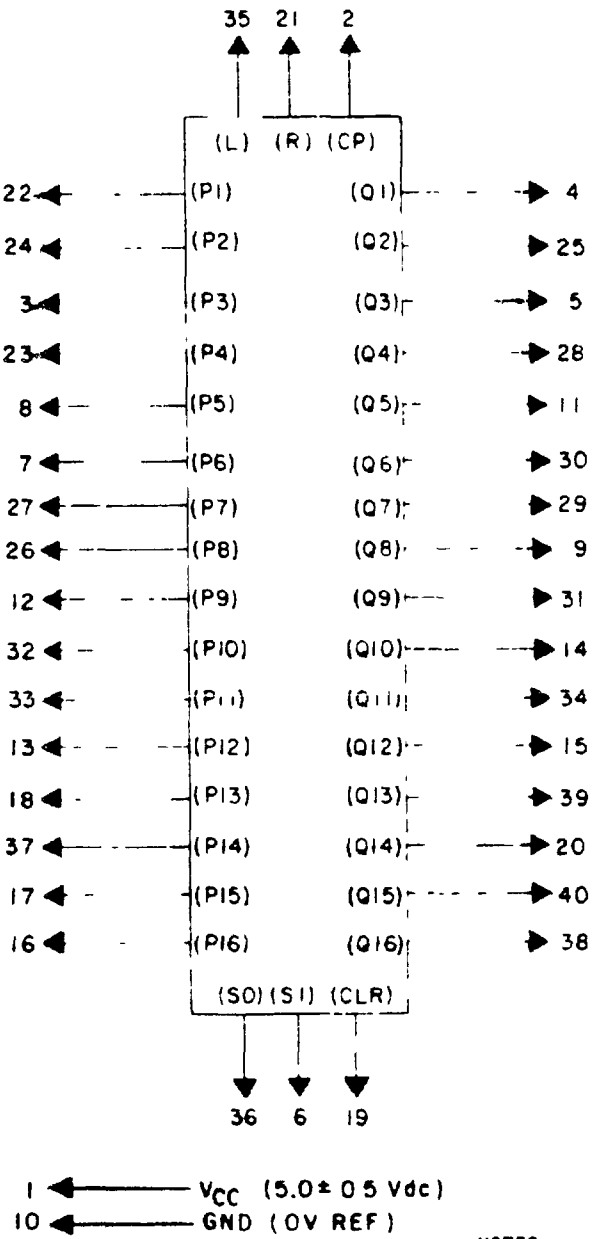
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES
1. P1-P16 = Parallel inputs
2. Q1-Q16 = Outputs
3. R = Shift right serial input
4. L = Shift left serial input
5. S0, S1 = Mode control inputs
6. CP = Clock input
7. CLR = Clear

FIGURE 1. CHF module function diagram, shift register

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	-	1	-	10	-	1	-	MΩ
B1-B16	V_{OUT} (low loaded)	IV	S	10	700	10	700	10	700	10	700	mV
C1-C18	Input breakdown current	V	S	0	100	0	100	0	100	0	100	µA
C19-C22	Input breakdown current	V	S	0	400	0	400	0	400	0	400	µA
D1-D16	Output short circuit current	VI	X	-6	-135	-6	-135	-6	-135	-6	-135	mA
E1-E16	Output leakage current	VII	X	0	250	0	250	0	250	0	250	µA
F1-F90	Functional	VIII	X	Test vectors must produce output vectors as specified in table VIII.								
G1-G22	Input clamp diode voltage	IX	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H16	Static output voltage, low level	X	X	0	500	0	500	0	500	0	500	mV
J1-J16	Static output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K18	Input current, high level	XII	X	0	20	0	20	0	20	0	20	µA
K19-K22	Input current, high level	XII	X	0	80	0	80	0	80	0	80	µA
L1-L18	Input current, low level	XIII	X	-10	-430	-10	-430	-10	-430	-10	-430	µA
L19-L21	Input current, low level	XIII	X	-0.1	-1.58	-0.1	-1.58	-0.1	-1.58	-0.1	-1.58	mA
L22	Input current, low level	XIII	X	-0.1	-1.94	-0.1	-1.94	-0.1	-1.94	-0.1	-1.94	mA
M	Power supply current	XIV	X	10	92	10	92	10	92	10	92	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF	XV	S	10	50	10	50	10	55	10	55	ns

See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N19-N36	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	10	80	10	80	10	90	10	90	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF	XVI	S	3	18	3	18	3	20	3	20	ns
P19-P36	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	6	40	6	40	6	40	6	40	ns
Q1-Q18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	S	5	27	5	27	5	30	5	30	ns
Q19-Q36	Propagation delay time (t_{PLH}), C_L = 150 pF	XVII	Q	5	32	5	32	5	35	5	35	ns
R1-R18	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	5	31	5	31	5	34	5	34	ns
R19-R36	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	-	41	5	41	5	44	5	44	ns
S1-S18	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	35	5	35	5	38	5	38	ns
S19-S36	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	5	45	5	45	5	48	5	48	ns
T1, T3	Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q1 and Q8 output voltage, high level)	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T2, T4	Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q16 and Q9 output voltage, low level)	XX	S	0.0	0 5	0.0	0 5	0.0	0.5	0 0	0 5	V
U1-U18	Maximum operating frequency check (t_{PW})	XXI	S	112	138	112	138	112	138	112	138	ns
V1-V16	Output crosstalk, low level	XXII	Q	0	650	0	650	0	650	0	650	mV
V17-V32	Output crosstalk, low level duration	XXII	Q	0	400	0	400	0	400	0	400	mV
W1-W16	Output crosstalk, high level	XXI.	Q	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XXIII	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply filter capacitance	XXIV	S	1 6	3 0	1.6	3 0	1 6	3 0	1 6	3 0	µF
Z1-Z18	Input capacitance	XXV	Q	2	20	2	20	2	20	2	20	pF
Z19-Z22	Input capacitance	XXV	Q	2	70	2	70	2	70	2	70	pF
AA1-AA48	Noise sensitivity (CLR, S0, S1)	XXVI	Q	450	550	450	550	450	550	450	550	ns
AA49-AA64	Noise sensitivity (CP)	XXVI	Q	0 0	1 5	0 0	1 5	0 0	1 5	0 0	1.5	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only

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4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4 5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Y shall be performed in the order specified in 4 4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 5 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Z shall be performed in the order specified in 4.4 2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8)

6.3 Cross reference of classification. Part number M28787/71-1 supersedes part number M28787/71. The two parts are interchangeable.

* 6 4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	440 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0 5 V minimum, 0.7 V maximum
High level	2 0 V minimum, V_{CC} maximum

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Input current, P1-P16, R, L		
Low level	(V_{IL} = 0.0 V)	-0.43 mA maximum
	(V_{IL} = 0.4 V)	-0.36 mA maximum
High level	(V_{IH} = 2.5 V)	20.0 µA maximum
	(V_{IH} = 5.5 V)	0.1 mA maximum
Input current, S0, S1, CLR		
Low level	(V_{IL} = 0.0 V)	-1.58 mA maximum
	(V_{IL} = 0.4 V)	-1.44 mA maximum
High level	(V_{IH} = 2.5 V)	80.0 µA maximum
	(V_{IH} = 5.5 V)	0.4 mA maximum
Input current, CP		
Low level	(V_{IL} = 0.0 V)	-1.94 mA maximum
	(V_{IL} = 0.4 V)	-1.76 mA maximum
High level	(V_{IH} = 2.5 V)	80.0 µA maximum
	(V_{IH} = 5.5 V)	0.4 mA maximum
Output voltage		
Low level	(I_{OL} = 6 mA)	0.5 V maximum
High level	(I_{OH} = -400 µA)	2.5 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.5 V)	6 mA
High level	(V_{OH} = 2.5 V)	-400 µA
Setup time		
t_{SETUP} (P1-P16, R, L)		20 ns minimum
t_{SETUP} (S0, S1, CLR)		30 ns minimum
Maximum operating frequency		
f_{MAX} (50 pF load)		16 MHz
Pulse width		
t_{PW} (CP)		20 ns minimum
t_{PW} (CLR)		25 ns minimum
Hold time		
t_{HOLD} (P1-P16, R, L)		5 ns minimum
t_{HOLD} (S0, S1, CLR)		0 ns minimum

6.5 Module function. This module contains a 16-bit universal shift register. The register has four distinct modes of operation

a. Parallel (broadside)
b. Shift right (in the direction Q1 → Q16)
c. Shift left (in the direction Q16 → Q1)
d. Inhibit clock (do nothing)

Synchronous parallel loading is accomplished by applying the 16 bits of data to P1-P16 inputs and taking both mode control inputs (S0 and S1) high. The data is loaded in and appears at the outputs after the positive transition of the clock input. During loading serial data flow is inhibited.

Shift right is accomplished synchronously with the rising edge of the clock when S0 is high and S1 is low. Serial data for this mode is entered at the shift-right serial input (R). When S0 is low and S1 is high data shifts left synchronously and new data is entered at the shift-left serial input (L).

Clocking of the register is inhibited when both mode control inputs are low.

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6.5.1 Input to output relationship. The 16-bit bidirectional universal shift register has the input to output relationship of table II.

6.5.1.1 Module crosstalk relationship. When connected in the worst case mode of operation, all but one module output switching simultaneously from a low level to a high level, the CHF module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as 650 mV positive going spike coincident with the positive edge of the switching outputs. The crosstalk spike must settle out to less than 400 mV after a delay of 30 ns maximum referenced from the positive going edge of the clock pulse.

6.5.2 Parallel outputs Paralleling of outputs is not permissible.

6.6 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue

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TABLE II Input to output relationship 1/-7/

Function	Inputs							Outputs																
	Clear	Mode		Clock	Serial		Parallel																	
	CLR	S1	S0	CP	L	R	P1-P16	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16
Clear	L	X	X	X	X	X	X-X	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
	H	X	X	L	X	X	X-X	$Q1_0$	$Q2_0$	$Q3_0$	$Q4_0$	$Q5_0$	$Q6_0$	$Q7_0$	$Q8_0$	$Q9_0$	$Q10_0$	$Q11_0$	$Q12_0$	$Q13_0$	$Q14_0$	$Q15_0$	$Q16_0$
Parallel load	H	H	H	L-H	X	X	p1-p16	p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	p11	p12	p13	p14	p15	p16
Right shift	H	L	H	L-H	X	H	X-X	H	$Q1_n$	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$
	H	L	H	L-H	X	L	X-X	L	$Q1_n$	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$
Left shift	H	H	L	L-H	H	X	X-X	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$	$Q16_n$	H
	H	H	L	L-H	L	X	X-X	$Q2_n$	$Q3_n$	$Q4_n$	$Q5_n$	$Q6_n$	$Q7_n$	$Q8_n$	$Q9_n$	$Q10_n$	$Q11_n$	$Q12_n$	$Q13_n$	$Q14_n$	$Q15_n$	$Q16_n$	L
Inhibit	H	L	L	X	X	X	X-X	$Q1_0$	$Q2_0$	$Q3_0$	$Q4_0$	$Q5_0$	$Q6_0$	$Q7_0$	$Q8_0$	$Q9_0$	$Q10_0$	$Q11_0$	$Q12_0$	$Q13_0$	$Q14_0$	$Q15_0$	$Q16_0$

NOTES

1/ H = High level (steady state)

2/ L = Low level (steady state).

3/ X = Irrelevant (any input, including transitions)

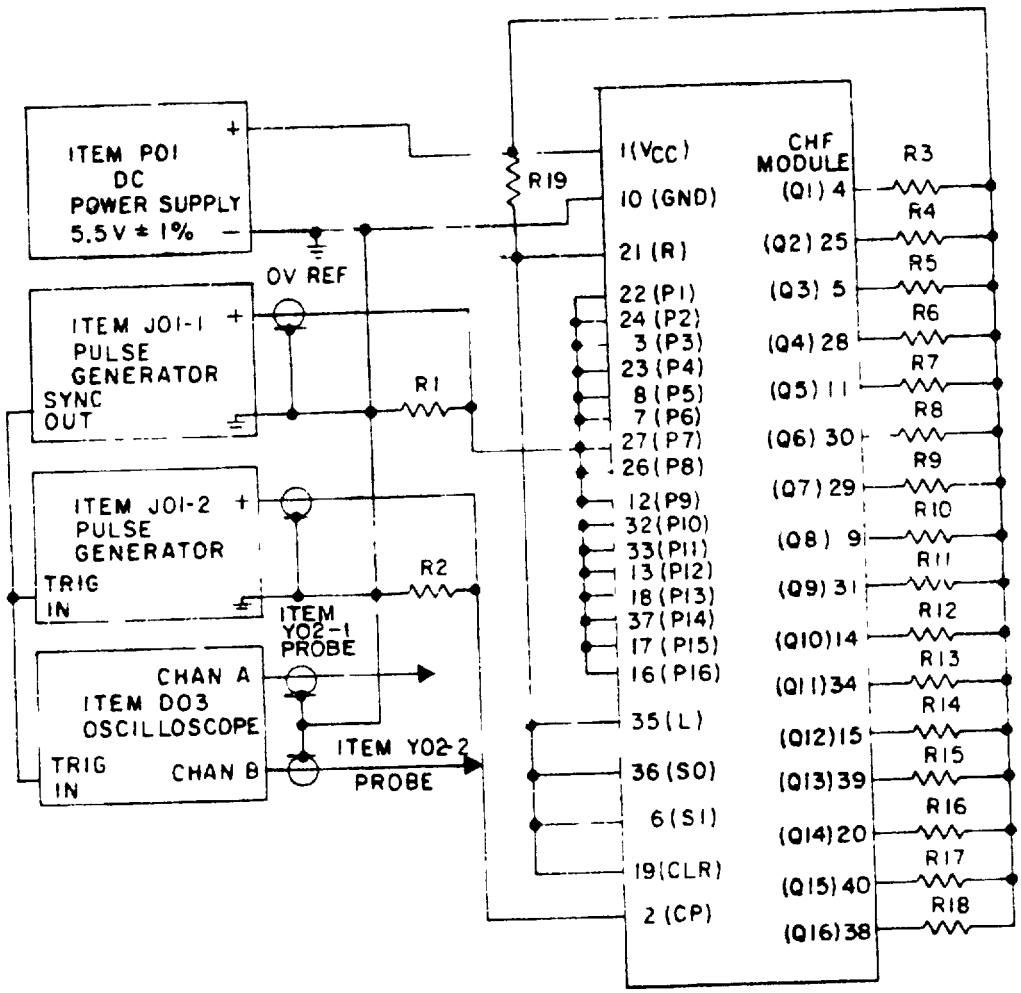
4/ L-H = Transition from low to high level.

5/ p1-p16 = The level of steady-state input at inputs P1-P16, respectively

6/ $Q1_0$-$Q16_0$ = The level of Q1-Q16, respectively, before the indicated steady-state input conditions were established

7/ $Q1_n$-$Q16_n$ = The level of Q1-Q16, respectively, before the most recent low to high transition of the clock.

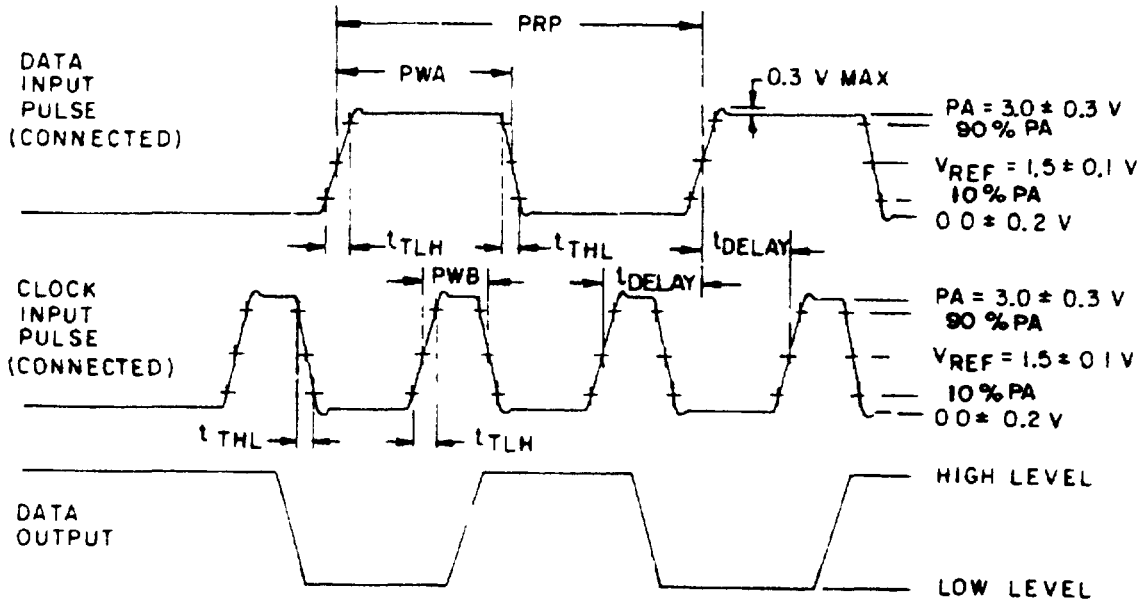
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NOTES:
1 R1 = R2 = 51Ω ±5%, 1/2 W
2 R3-R18 = 845Ω ±1%, 1/4 W
3 R19 = 1 kΩ ±5%, 1/2 W

FIGURE 2 Operating temperature and life test circuit

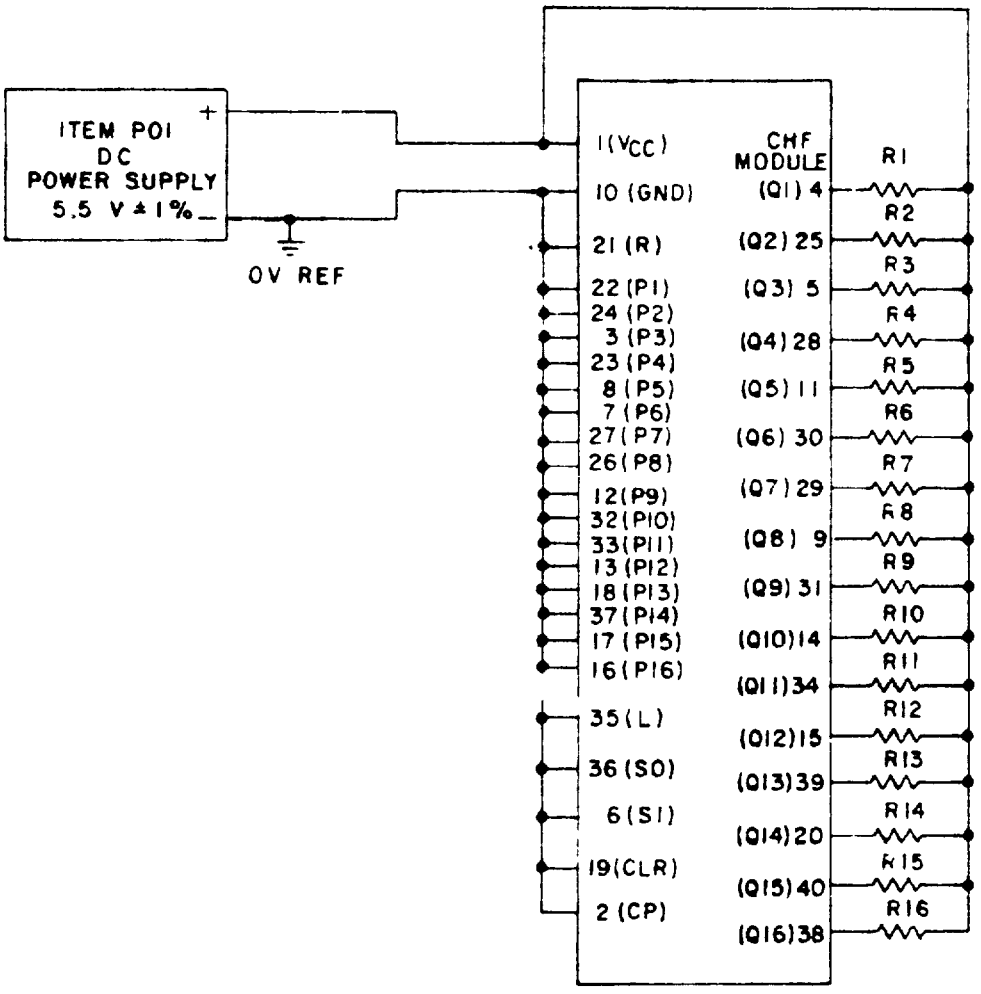
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NOTES
1 Input pulse parameters
$t_{TLH} = t_{THL} = 10 \pm 2$ ns
PRP = 1,000 ±200 ns
PWA = 500 ±100 ns
PWB = 100 ±20 ns
$t_{DELAY} = 250 \pm 50$ ns
2 Verify that the data output pulse high level ≥ 2.5 V and that the low level ≤ 0.5 V
3 All voltages are referenced to ground.

*FIGURE 3 Operating temperature and life test circuit waveforms for figure 2

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NOTE R1-R16 = 845Ω ±1%, 1/4 W

FIGURE 4 Component temperature test circuit

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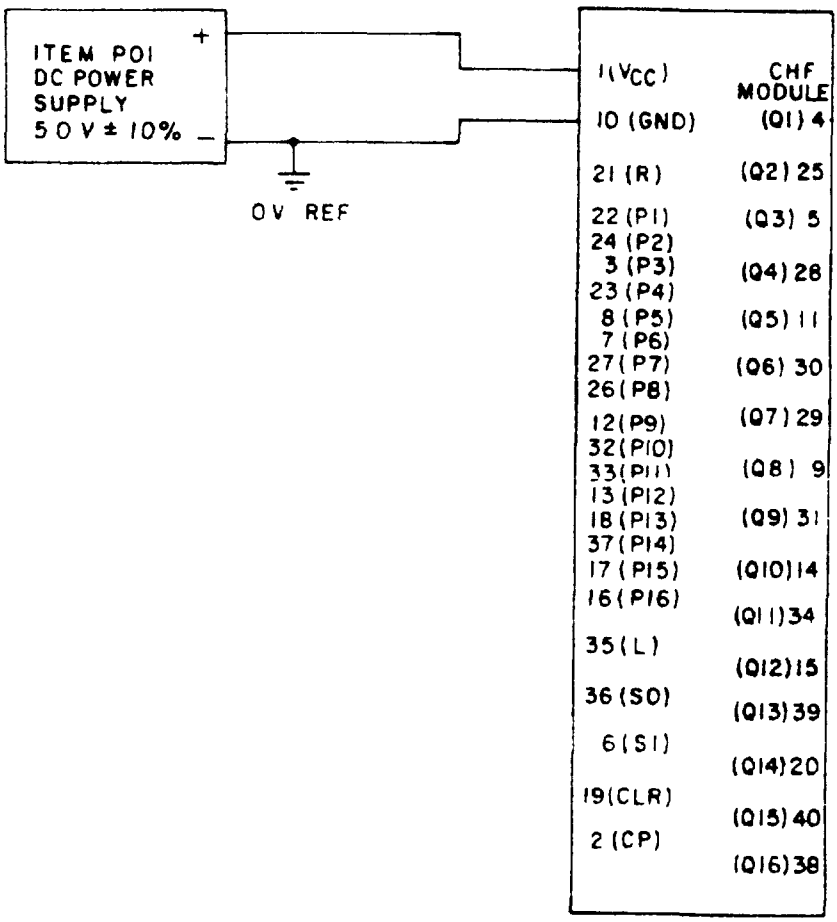


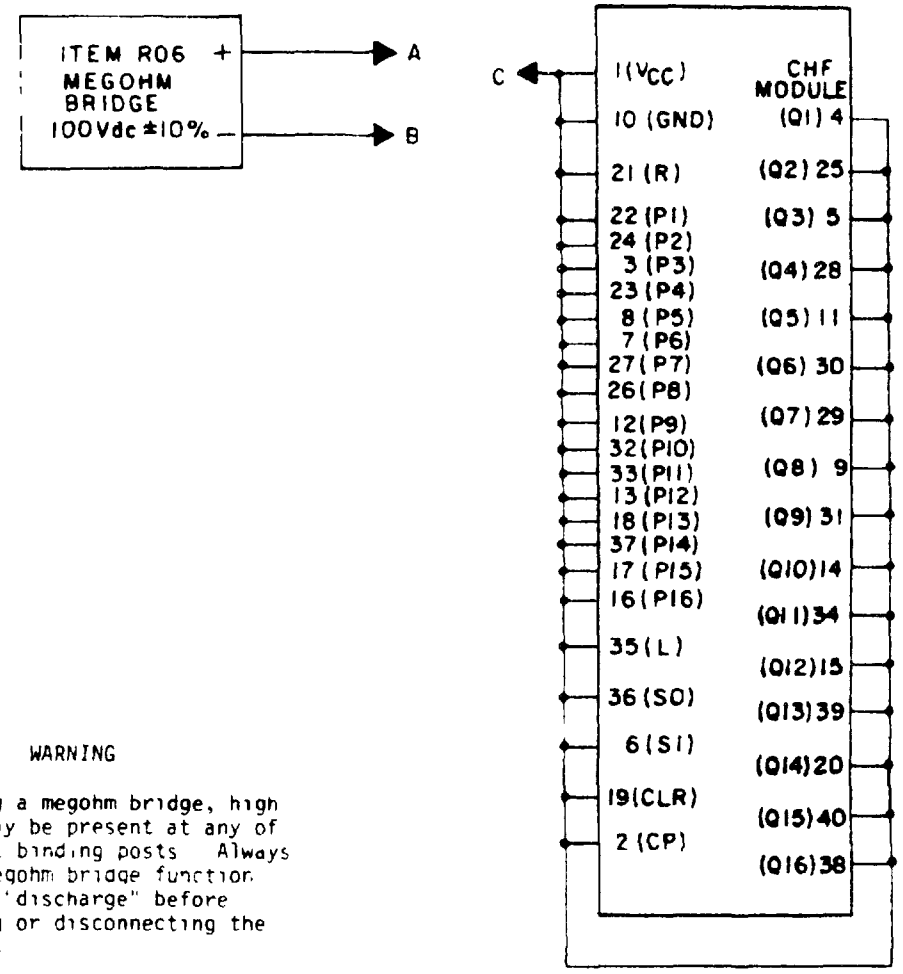
FIGURE 5 Vibration test circuit

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to 'discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

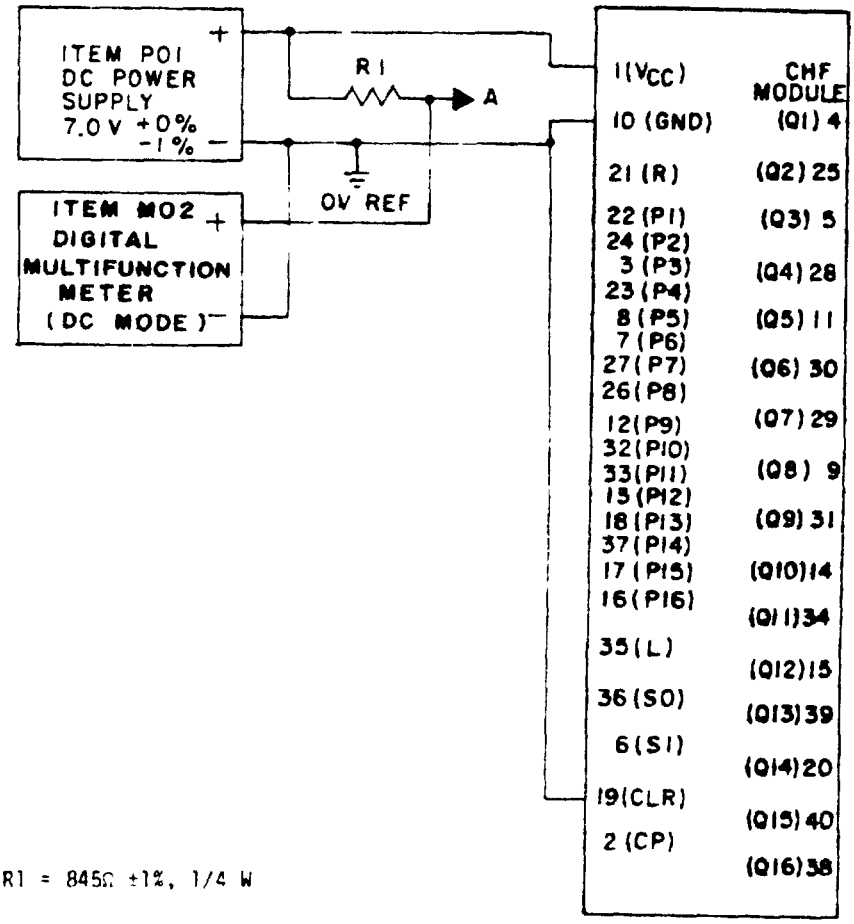
FIGURE 6 Isolation test circuit for table III

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* TABLE IV. V_{OUT} (low-loaded) test

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	Q1 V_{OUT} (low loaded)	4	10	700	mV	7
B2	Q2	25				
B3	Q3	5				
B4	Q4	28				
B5	Q5	11				
B6	Q6	30				
B7	Q7	29				
B8	Q8	9				
B9	Q9	31				
B10	Q10	14				
B11	Q11	34				
B12	Q12	15				
B13	Q13	39				
B14	Q14	20				
B15	Q15	40				
B16	Q16	38				

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NOTE R1 = 845Ω ±1%, 1/4 W

FIGURE 7 V_{OUT} (low-loaded) test circuit for table IV

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* TABLE V. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	R Input breakdown current	21	2,3,6,7,8,12, 13,16,17,18, 19,22,23,24, 26,27,32,33, 35,36,37	0	100	µA	8
C2	P1	22	2,3,6,7,8,12, 13,16,17,18, 19,21,23,24, 26,27,32,33, 35,36,37				
C3	P2	24	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 26,27,32,33, 35,36,37				
C4	P3	3	2,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36,37				
C5	P4	23	2,3,6,7,8,12, 13,16,17,18, 19,21,22,24, 26,27,32,33, 35,36,37				
C6	P5	8	2,3,6,7,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36,37				

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* TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C7	P6 Input breakdown current	7	2,3,6,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36,37	0	100	µA	8
C8	P7	27	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,32,33, 35,36,37				
C9	P8	26	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,27,32,33, 35,36,37				
C10	P9	12	2,3,6,7,8,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 35,36,37				
C11	P10	32	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,33, 35,36,37				
C12	P11	33	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 35,36,37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
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* TABLE V. Input breakdown current test - Continued

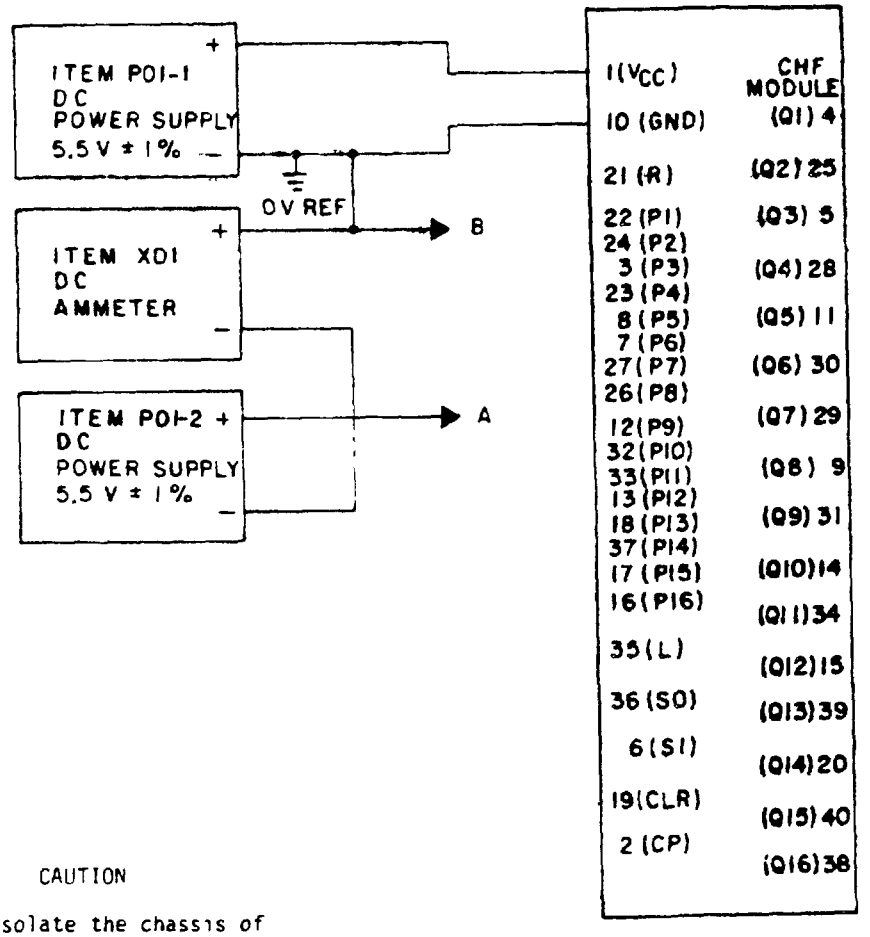
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C13	P12 Input breakdown current	13	2,3,6,7,8,12, 16,17,18,19, 21,22,23,24, 26,27,32,33, 35,36,37	0	100	µA	8
C14	P13	18	2,3,6,7,8,12, 13,16,17,19, 21,22,23,24, 26,27,32,33, 35,36,37				
C15	P14	37	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36				
C16	P15	17	2,3,6,7,8,12, 13,16,18,19, 21,22,23,24, 26,27,32,33, 35,36,37				
C17	P16	16	2,3,6,7,8,12, 13,17,18,19, 21,22,23,24, 26,27,32,33, 35,36,37				
C18	L	35	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,36,37				

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* TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	SO Input breakdown current	36	2,3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,37	0	400	µA	8
C20	S1	6	2,3,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36,37				
C21	CLR	19	2,3,6,7,8,12, 13,16,17,18, 21,22,23,24, 26,27,32,33, 35,36,37				
C22	CP	2	3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,35,36,37				

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CAUTION

Do not isolate the chassis of ammeter XD1 from earth ground

NOTE Item P01-2 shall be current limited to 5.0 mA

FIGURE 8 Input breakdown current test circuit for table V

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TABLE VI. Output short circuit current test

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Q1 Output short circuit current	22	4	3,7,8,12, 13,16,17, 18,23,24, 26,27,32, 33,37	-6	-135	mA	9, 10
D2	Q2	24	25	3,7,8,12, 13,16,17, 18,22,23, 26,27,32, 33,37				
D3	Q3	3	5	7,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37				
D4	Q4	23	28	3,7,8,12, 13,16,17, 18,22,24, 26,27,32, 33,37				
D5	Q5	8	11	3,7,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37				
D6	Q6	7	30	3,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37				

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TABLE VI Output short circuit current test - Continued.

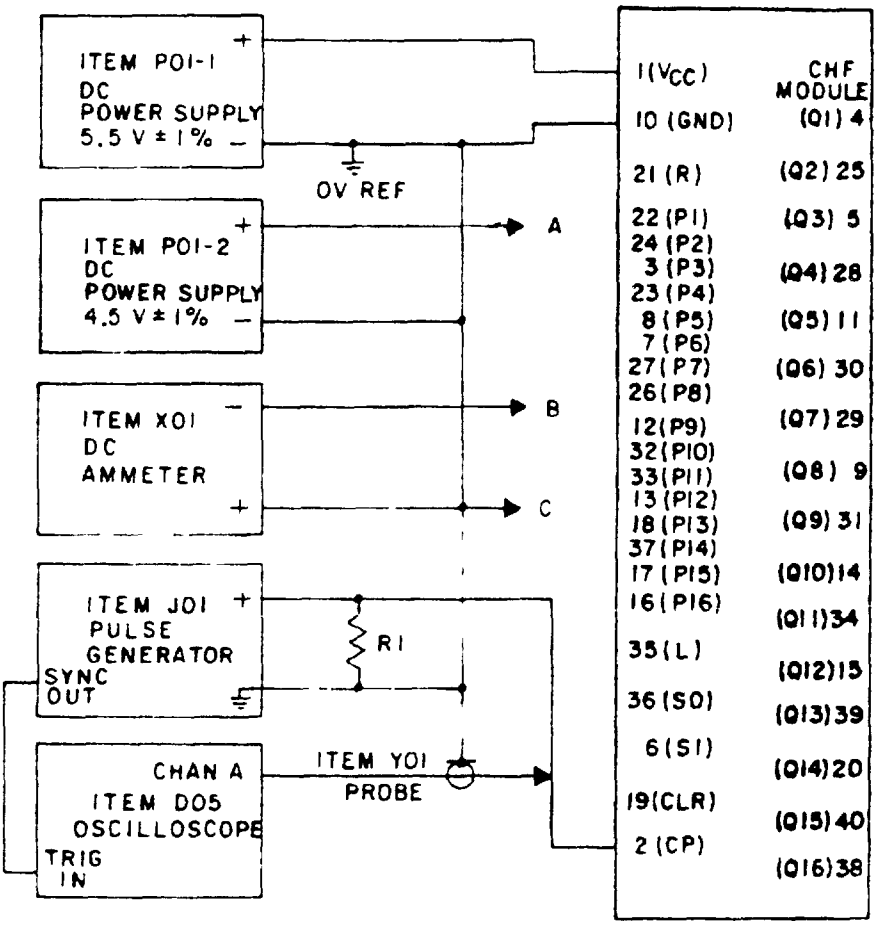
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D7	Q7 Output short circuit current	27	29	3,7,8,12, 13,16,17, 18,22,23, 24,26,32, 33,37	-6	-135	mA	9, 10
D8	Q8	26	9	3,7,8,12, 13,16,17, 18,22,23, 24,27,32, 33,37				
D9	Q9	12	31	3,7,8,13, 16,17,18, 22,23,24, 26,27,32, 33,37				
D10	Q10	32	14	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 33,37				
D11	Q11	33	34	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,37				
D12	Q12	13	15	3,7,8,12, 16,17,18, 22,23,24, 26,27,32, 33,37				

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TABLE VI. Output short circuit current test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D13	Q13 Output short circuit current	18	39	3,7,8,12, 13,16,17, 22,23,24, 26,27,32, 33,37	-6	-135	mA	9, 10
D14	Q14	37	20	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33				
D15	Q15	17	40	3,7,8,12, 13,16,18, 22,23,24, 26,27,32, 33,37				
D16	Q16	16	38	3,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,37				

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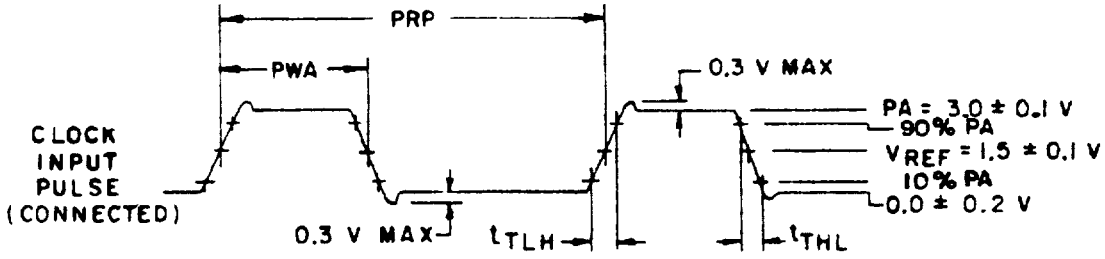
CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTES
1 R1 = 51Ω ±5%, 1/2 W
2 After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement
3 When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second

FIGURE 9 Output short circuit current test circuit for table VI

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NOTE Input pulse parameters
$t_{TLH} = t_{THL} = 25 \pm 10$ ns
PWA = 250 ±25 ns
PRP = 2,000 ±200 ns

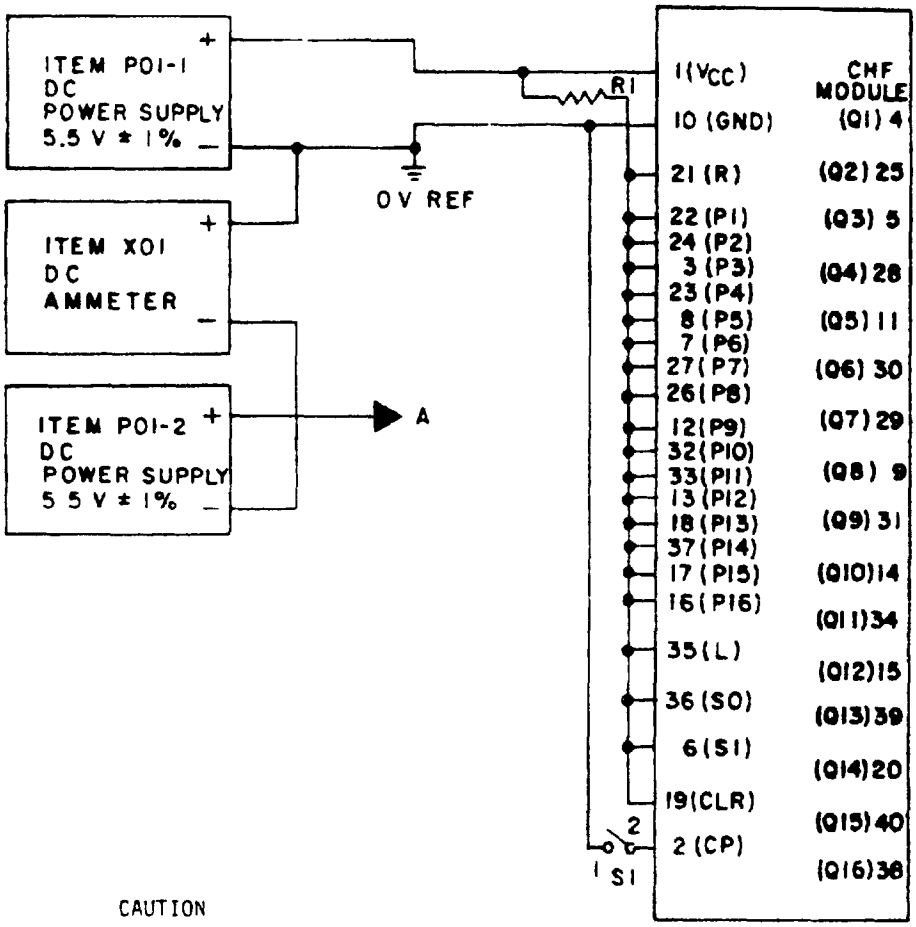
FIGURE 10 Output short circuit current test circuit waveform for table VI

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* TABLE VII. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	Q1 Output leakage current	4	0	250	µA	11
E2	Q2	25				
E3	Q3	5				
E4	Q4	28				
E5	Q5	11				
E6	Q6	30				
E7	Q7	29				
E8	Q8	9				
E9	Q9	31				
E10	Q10	14				
E11	Q11	34				
E12	Q12	15				
E13	Q13	39				
E14	Q14	20				
E15	Q15	40				
E16	Q16	38				

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

NOTES
1 R1 = 1 kΩ ±5%, 1/2 W
2 Place switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2, connect the ammeter and make the measurements
3 Item PO1-2 shall be current limited to 5 0 mA

FIGURE 11 Output leakage current test circuit for table VII

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TABLE VIII Functional test.

Test condition	Parameter	Connect				Output logic states															Initial 25° ±5°C ambient limits		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F1	Functional	3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37	2,19	2/,3/	Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	1/			12, 13
F2		3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	2		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F3		3,6,7,8,12,13,16,17,18,19,21,23,24,26,27,32,33,36,37	22,35		2	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F4		3,6,7,8,12,13,16,17,18,19,22,23,26,27,32,33,36,37	21,24,35			H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F5		6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	3,21,35			H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H				
F6		3,6,7,8,12,13,16,17,18,19,22,24,26,27,32,33,36,37	21,23,35			H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H				
F7		3,6,7,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	8,21,35			H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H				
F8		3,6,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	7,21,35			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H				
F9		3,6,7,8,12,13,16,17,18,19,22,23,24,26,32,33,36,37	21,27,35			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H				
F10		3,6,7,8,12,13,16,17,18,19,22,23,24,27,32,33,36,37	21,26,35			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H				
F11		3,6,7,8,13,16,17,18,19,22,23,24,26,27,32,33,36,37	12,21,35			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H				
F12		3,6,7,8,12,13,16,17,18,19,22,23,24,26,27,33,36,37	21,32,35			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H				
F13		3,6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,36,37	21,33,36			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H				
F14		3,6,7,8,12,16,17,18,19,22,23,24,26,27,32,33,36,37	13,21,35			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H				
F15		3,6,7,8,12,13,16,17,19,22,23,24,26,27,32,33,36,37	18,21,36			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H				
F16		3,6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36	21,35,37			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H				

See footnotes at end of table

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F17	Functional	3,6,7,8,12, 13,16,18,19, 22,23,24,26, 27,32,33,36, 37	17,21,35	2/,3/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	1/			12, 13
F18		3,6,7,8,12, 13,17,18,19, 22,23,24,26, 27,32,33,36, 37	16,21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L				
F19		6,16,19,21, 36	3,7,8,12,13, 17,18,22,23, 24,26,27,32, 33,35,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F20		6,17,19,21, 35,36	3,7,8,12,13, 16,18,22,23, 24,26,27,32, 33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L				
F21		6,19,21,35, 36,37	3,7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F22		6,18,19,21, 35,36	3,7,8,12,13, 16,17,22,23, 24,26,27,32, 33,37			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F23		6,13,19,21, 35,36	3,7,8,12,16, 17,18,22,23, 24,26,27,32, 33,37			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F24		6,19,21,33, 35,36	3,7,8,12,13, 16,17,18,22, 23,24,26,27, 32,37			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				
F25		6,19,21,32, 35,36	3,7,8,12,13, 16,17,18,22, 23,24,26,27, 33,37			L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L				
F26		6,12,19,21, 35,36	3,7,8,13,16, 17,18,22,23, 24,26,27,32, 33,37			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				
F27		6,19,21,26, 35,36	3,7,8,12,13, 16,17,18,22, 23,24,27,32, 33,37			L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L				
F28		6,19,21,27, 35,36	3,7,8,12,13, 16,17,18,22, 23,24,26,32, 33,37			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F29		6,7,19,21, 35,36	3,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,37			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F30	Functional	6,8,19,21, 35,36	3,7,12,13, 16,17,18,22, 23,24,26,27, 32,33,37	2/,3/	2	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L		1/		12, 13
F31		6,19,21,23, 35,36	3,7,8,12,13, 16,17,18,22, 24,26,27,32, 33,37			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F32		3,6,19,21, 35,36	7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,37			L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L				
F33		6,19,21,24, 35,36	3,7,8,12,13, 16,17,18,22, 23,26,27,32, 33,37			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F34		6,19,21,22, 35,36	3,7,8,12,13, 16,17,18,23, 24,26,27,32, 33,37			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F35		6,19,21,35, 36	3,7,8,12,13, 16,17,18,22, 23,24,26,27, 32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F36		3,7,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 35,37	6,36			H	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F37		3,6,7,8,12, 13,16,17,18, 19,22,23,24, 26,27,32,33, 36,37	21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F38		19	3,6,7,8,12, 13,16,17,18, 21,22,23,24, 26,27,32,33, 35,36,37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F39		19,22,36	3,6,7,8,12, 13,16,17,18, 21,23,24,26, 27,32,33,35, 37			L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F40		19,21,24,36	3,6,7,8,12, 13,16,17,18, 22,23,26,27, 32,33,35,37			H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H				

See footnotes at end of table

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F41	Functional	3,19,21 3	6,7,8,12,13, 16,17,18 22, 23 24 26,27, 32,33,35 37	2/,3/	2	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	1/		12, 13
F42		19,21,23,36	3,6,7,8,12, 13,16,17,18, 22,24,26,27, 32,33,35,37			H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H			
F43		8,19,21,36	3,6,7,12,13, 16,17,18,22, 23 24,26,27, 32,33,35,37			H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H			
F44		7,19,21,36	3,6,8,12 13, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H			
F45		19,21,27,36	3,6,7,8,12, 13,16,17,18, 22,23,24 26, 32,33,35,37			H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H			
F46		19,21,26,36	3,6,7,8,12, 13,16,17,18, 22,23,24 27 32,33,35,37			H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H			
F47		12,19,21,36	3,6,7,8,13, 16,17,18 22, 23,24 26 27, 32,33,35,37			H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H			
F48		19,21,32,36	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 27,33,35,37			H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H			
F49		19,21,33,36	3,6,7,8,12, 13,16,17,18 22,23,24,26, 27,32,35,37			H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H			
F50		13,19,21,36	3,6,7,8,12, 16,17,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H			
F51		18,19,21 36	3,6,7,8,12, 13,16,17,22, 23,24,26,27, 32,33 35,37			H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H			
F52		19,21,36,37	3,6,7,8,12, 13,16,17,18, 22,23,24,26, 27,32,33,35			H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H			
F53		17,19,21,36	3,6,7,8,12, 13,16,18,22, 23,24,26,27, 32,33,35,37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H			

See footnotes at end of table

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F54	Functional	16,19,21,36	3,6,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,35,37	2/,3/	2	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	1/			12, 13
F55		19,21,36	3,6,7 8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,35, 37			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F56		6,19,21,35,36	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F57		3,6,7,8,12, 13,17,18,19, 21,22,23,24, 26,27,32,33, 35,37	16,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F58		3,6,7,8,12, 13,16,18,19, 21,22,23,24, 26,27,32,33, 37	17,35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	L				
F59		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33	35,36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	H	L	L				
F60		3,6,7,8,12, 13,16,17,19, 21,22,23,24, 26,27,32,33, 37	18,35,36			L	L	L	L	L	L	L	L	L	L	L	L	H	L	L	L				
F61		3,6,7,8,12, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	13,35,36			L	L	L	L	L	L	L	L	L	L	L	H	L	L	L	L				
F62		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 37	33,35,36			L	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L				

See footnotes at end of table

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F63	Functional	3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,33, 37	32 35 36	2/,3/	2	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	1/			12 13
F64		3,6,7,8,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	12,35,36			L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L				
F65		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,27,32,33, 37	26,35,36			L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L				
F66		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,32,33, 37	27,35,36			L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L				
F67		3,6,8,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	7,35,36			L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	L				
F68		3,6,7,12,13, 16,17,18,19, 21,22,23,24, 26,27,32,33, 37	8,35,36			L	L	L	L	H	L	L	L	L	L	L	L	L	L	L	L				
F69		3,6,7,8,12, 13,16,17,18, 19,21,22,24, 26,27,32,33 37	23,35,36			L	L	L	H	L	L	L	L	L	L	L	L	L	L	L	L				
F70		6,7,8,12,13, 16,17,18 19, 21,22,23,24, 26,27,32,33, 37	3,35,36			L	L	H	L	L	L	L	L	L	L	L	L	L	L	L	L				
F71		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 26,27,32,33, 37	24,35,36			L	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F72		3,6,7,8,12, 13,16,17,18, 19,23,24,26, 27,32,33,37	21,22,35,36			H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F73		3,6,7,8,12, 13,16,17,18, 19,21,22,23, 24,26,27,32, 33,37	35,36			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				

See footnotes at end of table

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TABLE VIII Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F74	Functional	3,6,8,12,17, 18,19,22,27, 33,35,36	7,13,16,21, 23,24,26, 32,37	2/,3/	2	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	1/			12, 13
F75		3,8,12,17,18, 19 22,27,33, 36	6,7,13,16, 21 23 24, 26,32,35 37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F76		3,8,12,17,18, 19 22,27,33, 35	6,7,13,16, 21,23,24, 26,32,36,37			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F77		6,7,13,16,19, 23,24,26,32, 37	3,8,12,17, 18,21,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F78		7,13,16,19, 21,23,24,26, 32,37	3,6,8,12, 17,18,22, 27,33,35,36			H	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L				
F79		6,7,13,16,19, 21,23,24,26, 32,36,37	3,8,12,17, 18,22,27, 33,35			L	H	L	H	L	H	L	H	L	H	L	H	L	H	L	H				
F80		3,6,8,12,17, 18,22,27,33, 35,36	7,13,16,19 21,23,24, 26,32,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F81		Open	2,3,6,7,8, 12,13,16, 17,18,19, 21,22,23, 24,26,27, 32,33,35, 36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F82		19	3,6,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 35,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F83		Open	2,3,6,7,8, 12,13,16, 17,18,19, 21,22 23, 24 26,27, 32,33,35, 36,37		Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F84		6,19,22,36	3,7,8,12, 13,16,17, 18,21,23, 24,26,27, 32,33,34, 35,37		2	H	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				
F85		6,19	3,7,8,12, 13,16,17, 18,21,22, 23,24,26, 27,32,33, 34,35,36, 37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				

See footnotes at end of table

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TABLE VIII Functional test - Continued

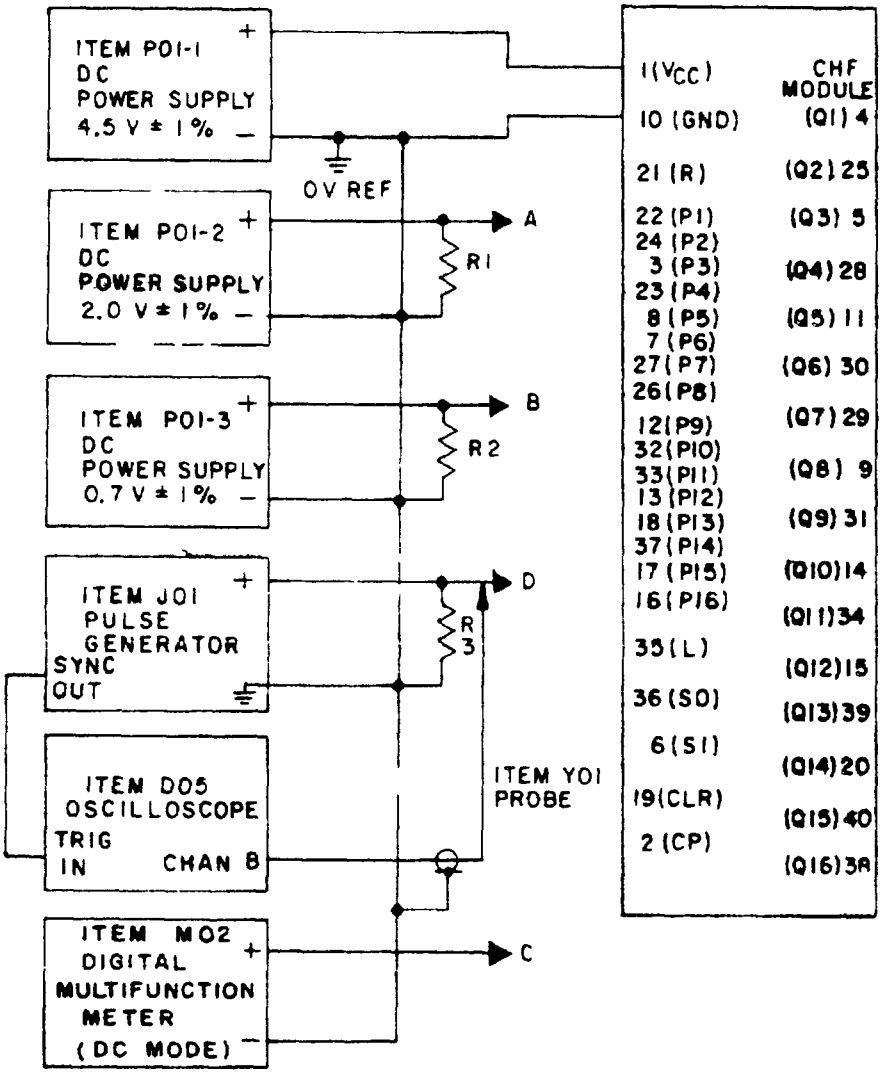
Test condition	Parameter	Connect				Output logic states																Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Min	Max		
F86	Functional	Open	2,3,6,7,8,12,13,16,17,18,19,21,22,23,24,26,27,32,33,35,36,37	2/,3/	Open	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	1/			12, 13
F87		6,19,35	3,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,36,37		2	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F88		19	3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H				
F89		3,6,7,8,12,13,16,17,18,19,22,23,24,26,27,32,33,36,37	21,35			H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H				
F90		3,6,7,8,12,13,16,17,18,21,22,23,24,26,27,32,33,35,36,37	19			L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L				

1/ Output logic states
L = Low level = 0.0 to 0.5 V
H = High level = 2.5 to 4.5 V.

2/ For test conditions F1-F90 verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished. Test conditions F1-F90 are to be performed in sequence.

3/ Q1-Q16 output logic states are measured at output pins 4, 25, 5, 28, 11, 30, 29, 9, 31, 14, 34, 15, 39, 20, 40, and 38 respectively.

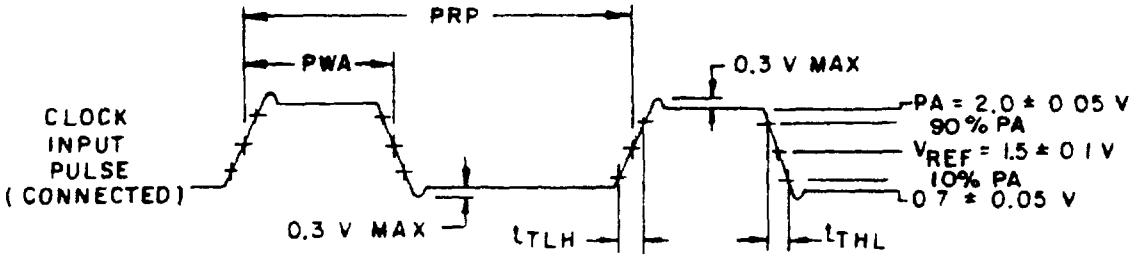
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NOTES
1. R1 - R3 = 51Ω ±5 , 1/2 W
2. After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply one pulse before making each measurement

FIGURE 12 Functional test circuit for table VIII

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NOTE Input pulse parameters
$t_{TLH} = t_{THL} = 25 \pm 10$ ns
PWA = 250 ±25 ns
PRP = 2,000 ±200 ns

FIGURE 13 Functional test circuit waveform for table VIII

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TABLE IX. Input clamp diode voltage test.

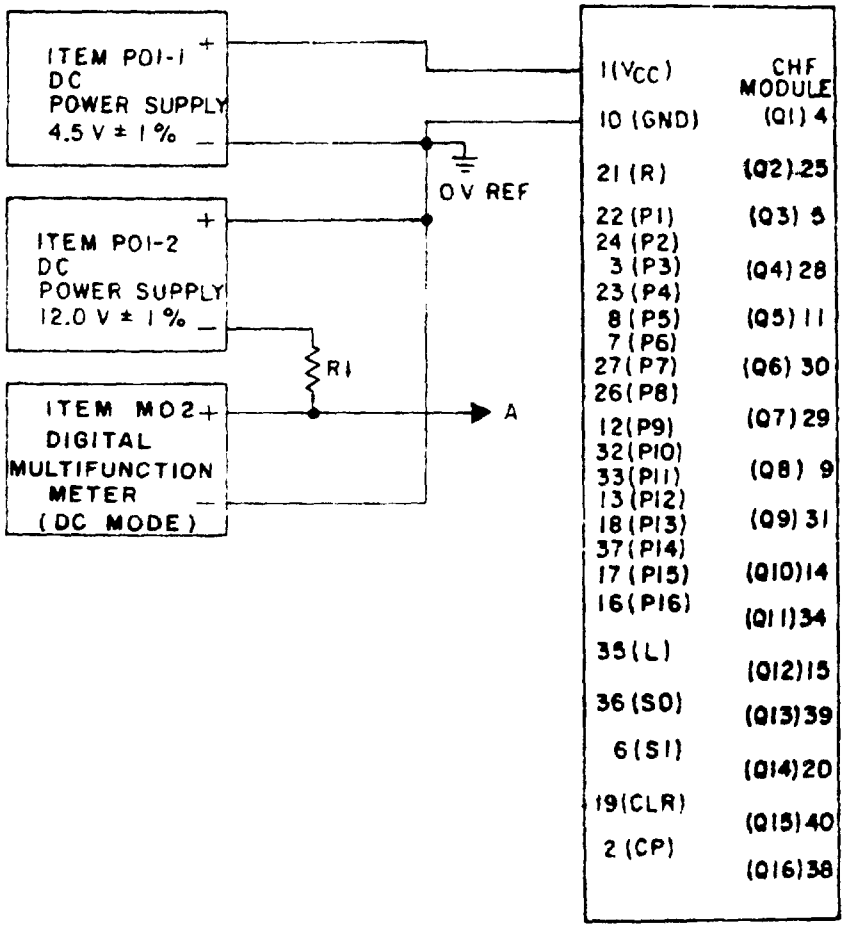
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	R Input clamp diode voltage	21	-0.4	-1.5	V	14
G2	P1	22				
G3	P2	24				
G4	P3	3				
G5	P4	23				
G6	P5	8				
G7	P6	7				
G8	P7	27				
G9	P8	26				
G10	P9	12				
G11	P10	32				
G12	P11	33				
G13	P12	13				
G14	P13	18				
G15	P14	37				
G16	P15	17				
G17	P16	16				
G18	L	35				
G19	SO	36				

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TABLE IX. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G20	S1 Input clamp diode voltage	6	-0.4	-1.5	V	14
G21	CLR Input clamp diode voltage	19	-0.4	-1 5	V	14
G22	CP Input clamp diode voltage	2	-0.4	-1.5	V	14

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NOTE R1 = 590Ω ±1%, 1/4 W

* FIGURE 14 Input clamp diode voltage test circuit for table I)

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* TABLE X. Static output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Q1 Static output voltage, low level	22	6,19,36	4	0	500	mV	15,16
H2	Q2	24		25				
H3	Q3	3		5				
H4	Q4	23		28				
H5	Q5	8		11				
H6	Q6	7		30				
H7	Q7	27		29				
H8	Q8	26		9				
H9	Q9	12		31				
H10	Q10	32		14				
H11	Q11	33		34				
H12	Q12	13		15				
H13	Q13	18		39				
H14	Q14	37		20				
H15	Q15	17		40				
H16	Q16	16		38				

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TABLE XI Static output voltage (high-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	Q1 Static output voltage, high level	3,7,8,12, 13,16,17, 18,21,23, 24,26,27, 32,33,35,37	6,19,22,36	4	2 5	4.5	V	15,16
J2	Q2	3,7,8,12, 13,16,17, 18,21,22, 23,26,27, 32,33,35,37	6,19,24,36	25				
J3	Q3	7,8,12,13, 16,17,18, 21,22,23, 24,26,27, 32,33,35, 37	3,6,19,36	5				
J4	Q4	3,7,8 2, 13,16, 7, 18,2 ,22, 24,26,27, 32,33,35, 37	6,19,23,26	28				
J5	Q5	3,7,12,13, 16,17,18, 21,22,23, 24,26,27, 32,33,35, 37	6,8,19,36	11				
J6	Q6	3,8,12,13, 16,17,18, 21,22,23, 24,26,27, 32,33,35, 37	6,7,19,36	30				
J7	Q7	3,7,8,12, 13,16,17, 18,21,22, 23,24,26, 32,33,35,37	6,19,27,36	29				

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TABLE XI. Static output voltage (high-level) test - Continued.

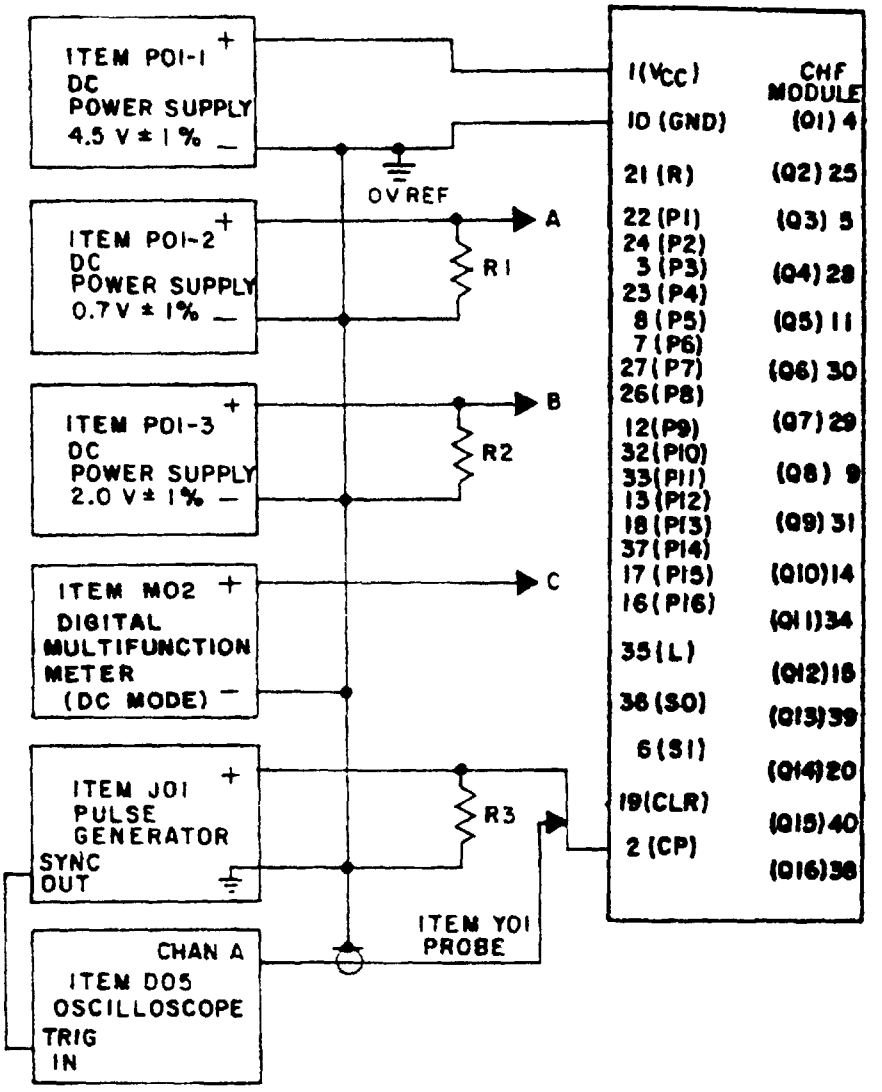
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J8	Q8 Static output voltage, high level	3,7,8,12,13, 16,17,18,21, 22,23,24,27, 32,33,35,37	6,19,26,36	9	2.5	4.5	V	15,16
J9	Q9	3,7,8,13,16, 17,18,21,22, 23,24,26,27, 32,33,35,37	6,12,19,36	31				
J10	Q10	3,7,8,12,13, 16,17,18,21, 22,23,24,26, 27,33,35,37	6,19,32,36	14				
J11	Q11	3,7,8,12,13, 16,17,18,21, 22,23,24,26, 27,32,35,37	6,19,33,36	34				
J12	Q12	3,7,8,12,16, 17,18,21,22, 23,24,26,27, 32,33,35,37	6,13,19,36	15				
J13	Q13	3,7,8,12,13, 16,17,21,22, 23,24,26,27, 32,33,35,37	6,18,19,36	39				
J14	Q14	3,7,8,12,13, 16,17,18,21, 22,23,24,26, 27,32,33,35	6,19,36,37	20				

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TABLE XI. Static output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J15	Q15 Static output voltage, high level	3,7,8,12,13, 16,18,21,22, 23,24,26,27, 32,33,35,37	6,17,19,36	40	2.5	4.5	V	15,16
J16	Q16 Static output voltage, high level	3,7,8,12,13, 17,18,21,22, 23,24,26,27, 32,33,35,37	6,16,19,36	38	2.5	4.5	V	15,16

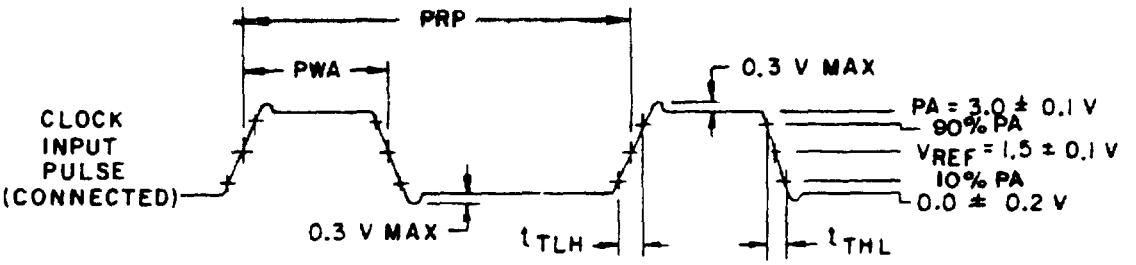
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NOTES
1. R1 - R3 = 51Ω ±5%, 1/2 W.
2 After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement
3 Load each output with a load as specified on figure 17.

FIGURE 15 Static output voltage (low-level and high-level) test circuit for tables X and XI

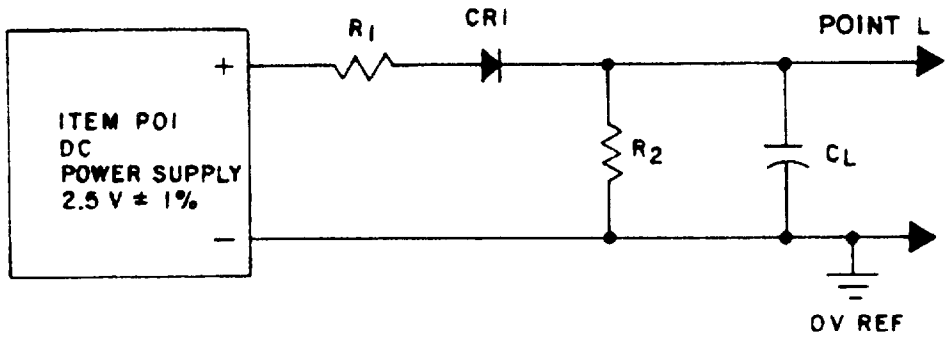
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NOTE Input pulse parameters.
$t_{TLH} = t_{THL} = 25 \pm 10$ ns
PWA = 250 ±25 ns
PRP = 2,000 ±200 ns.

FIGURE 16 Static output voltage (low-level and high-level) test circuit waveform for tables X and XI

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NOTES
1. CR1 = 1N3600, 1N4150, or equivalent.
* 2. R1 = Select value for a current flow of 6 mA ±5% out of point L with point L held at 0 5 V
3. R2 = 6 34 kΩ ±1%, 1/8 W
4. Test type S
   C_L = 50 ±5 pF (including probe and parasitic capacitance)
   Test type Q
   C_L = 150 ±5 pF (including probe and parasitic capacitance)
5. Capacitor C_L is not required for dc electrical test conditions H1-H16 and J1-J16
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25 4 mm)

FIGURE 17 CHF logic load circuit

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TABLE XII. Input current (high-level) test.

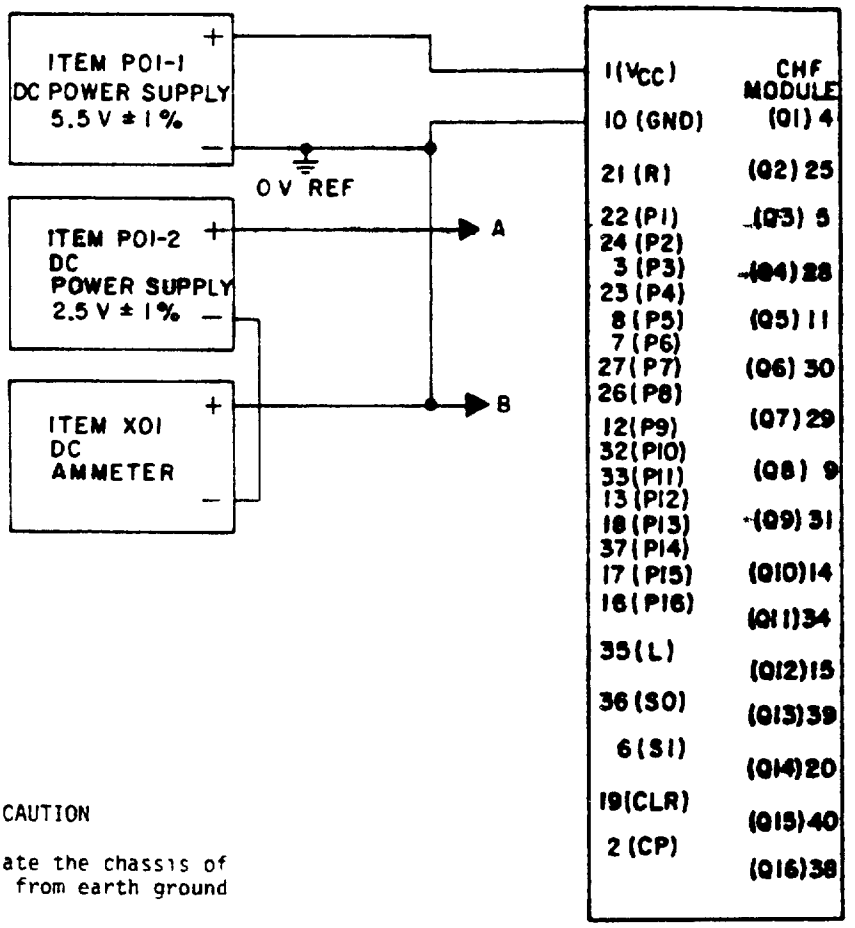
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	R Input current, high level	21	2,3,6,7,8,12,13,16,17, 18,19,22,23,24,26,27, 32,33,35,36,37	0	20	µA	18
K2	P1	22	2,3,6,7,8,12,13,16,17, 18,19,21,23,24,26,27, 32,33,35,36,37				
K3	P2	24	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,26,27, 32,33,35,36,37				
K4	P3	3	2,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36,37				
K5	P4	23	2,3,6,7,8,12,13,16,17, 18,19,21,22,24,26,27, 32,33,35,36,37				
K6	P5		2,3,6,7,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36,37				
K7	P6	7	2,3,6,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36,37				
K8	P7	27	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 32,33,35,36,37				
K9	P8	26	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,27, 32,33,35,36,37				
K10	P9	12	2,3,6,7,8,13,16,17,18, 19,21,22,23,24,26,27, 32,33,35,36,37				
K11	P10	32	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,13,35,36,37				
K12	P11	33	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,35,36,37				

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TABLE XII Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K13	P12 Input current, high level	13	2,3,6,7,8,12,16,17,18, 19,21,22,23,24,26,27, 32,33,35,36,37	0	20	µA	18
K14	P13	18	2,3,6,7,8,12,13,16,17, 19,21,22,23,24,26,27, 32,33,35,36,37				
K15	P14	37	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36				
K16	P15	17	2,3,6,7,8,12,13,16,18, 19,21,22,23,24,26,27, 32,33,35,36,37				
K17	P16	16	2,3,6,7,8,12,13,17,18, 19,21,22,23,24,26,27, 32,33,35,36,37				
K18	L	35	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,36,37				
K19	SO	36	2,3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,37		80		
K20	S1	6	2,3,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36,37				
K21	CLR	19	2,3,6,7,8,12,13,16,17, 18,21,22,23,24,26,27, 32,33,35,36,37				
K22	CP	2	3,6,7,8,12,13,16,17, 18,19,21,22,23,24,26, 27,32,33,35,36,37				

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

FIGURE 18 Input current (high-level) test circuit for table XII

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TABLE XIII. Input current (low-level) test.

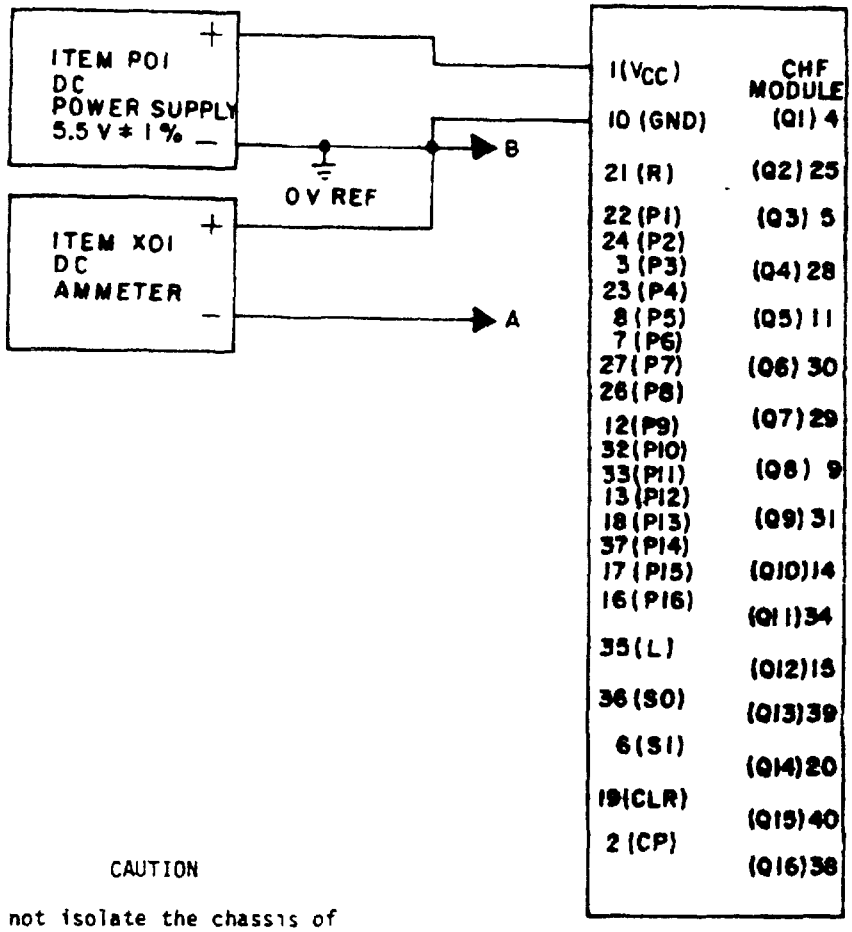
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	R Input current, low level	21	6	-10	-430	µA	19
L2	P1	22	Open				
L3	P2	24					
L4	P3	3					
L5	P4	23					
L6	P5	8					
L7	P6	7					
L8	P7	27					
L9	P8	26					
L10	P9	12					
L11	P10	32					
L12	P11	33					
L13	P12	13					
L14	P13	18					
L15	P14	37					
L16	P15	17					
L17	P16	16					
L18	L	35	36				
L19	SO	36	Open	-0.10	-1.58	mA	

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TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L20	S1 Input current, low level	6	Open	-0.10	-1.58	mA	19
L21	CLR Input current, low level	19	Open	-0.10	-1.58	mA	19
L22	CP Input current, low level	2	Open	-0.10	-1.94	mA	19

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

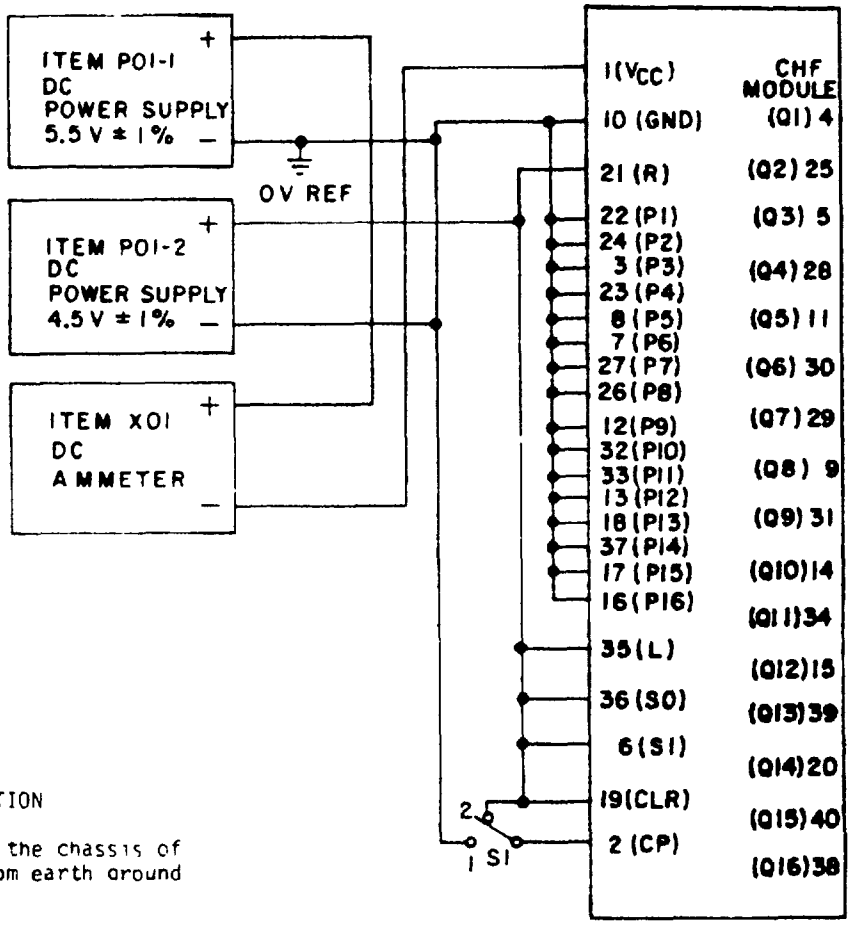
FIGURE 19 Input current (low-level) test circuit for table XIII

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	10	92	mA	20

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTES
1 S1 = SPDT switch
2 Momentarily put switch S1 to position 1
3 Switch S1 to position 2

* FIGURE 20 Power supply current test circuit for table XIV

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* TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
N1	Q1 Transition time (t_{TLH}), C_L = 50 pF	6, 36	Open	4	10	50	ns	21,22
N2	Q2			25				
N3	Q3			5				
N4	Q4			28				
N5	Q5			11				
N6	Q6			30				
N7	Q7			29				
N8	Q8			9				
N9	Q9			31				
N10	Q10			14				
N11	Q11			34				
N12	Q12			15				
N13	Q13			39				
N14	Q14			20				
N15	Q15			40				
N16	Q16			38				
N17	Q1	36	6	4				
N18	Q16	6	36	38				

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* TABLE XV Transition time (t_{TLH}) test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N19	Q1 Transition time (t_{TLH}), C_L = 150 pF	6, 36	Open	4	10	80	ns	21,22
N20	Q2			25				
N21	Q3			5				
N22	Q4			28				
N23	Q5			11				
N24	Q6			30				
N25	Q7			29				
N26	Q8			9				
N27	Q9			31				
N28	Q10			14				
N29	Q11			34				
N30	Q12			15				
N31	Q13			39				
N32	Q14			20				
N33	Q15			40				
N34	Q16			38				
N35	Q1	36	6	4				
N36	Q16	6	36	38				

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* TABLE XVI. Transition time (t_{THL}) test.

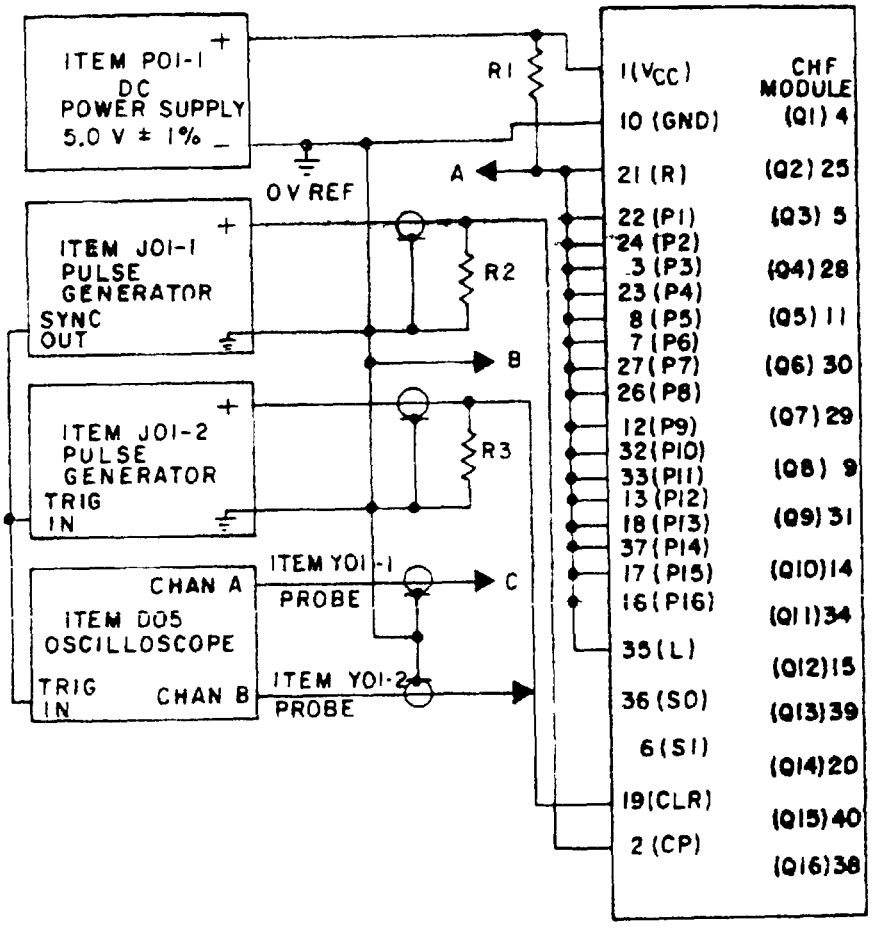
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q1 Transition time (t_{THL}), C_L = 50 pF	6, 36	Open	4	3	18	ns	21,22
P2	Q2			25				
P3	Q3			5				
P4	Q4			28				
P5	Q5			11				
P6	Q6			30				
P7	Q7			29				
P8	Q8			9				
P9	Q9			31				
P10	Q10			14				
P11	Q11			34				
P12	Q12			15				
P13	Q13			39				
P14	Q14			20				
P15	Q15			40				
P16	Q16			38				
P17	Q1	36	6	4				
P18	Q16	6	36	38				

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* TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P19	Q1 Transition time (t_{THL}), C_L = 150 pF	6, 36	Open	4	6	40	ns	21,22
P20	Q2			25				
P21	Q3			5				
P22	Q4			28				
P23	Q5			11				
P24	Q6			30				
P25	Q7			29				
P26	Q8			9				
P27	Q9			31				
P28	Q10			14				
P29	Q11			34				
P30	Q12			15				
P31	Q13			39				
P32	Q14			20				
P33	Q15			40				
P34	Q16			38				
P35	Q1	36	6	4				
P36	Q16	6	36	38				

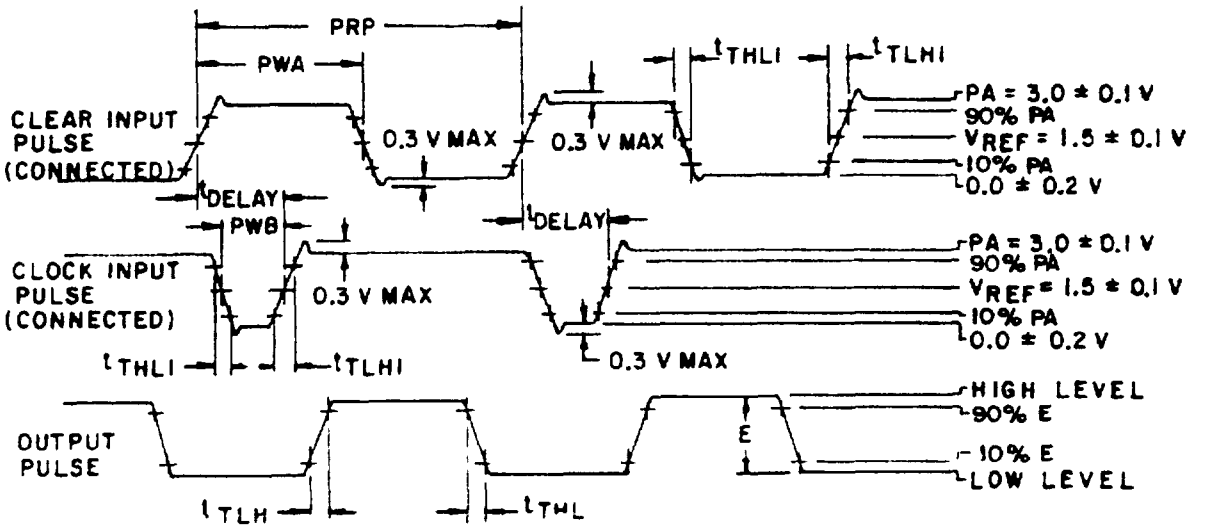
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NOTES
1 R1 = 1 kΩ ±1%, 1/4 W
2 R2 = R3 = 51Ω ±5%, 1/2 W
3 Load each output with a load as specified on figure 17

*FIGURE 21 Transition time (t_{TLH} and t_{THL}) test circuit for tables XV and XVI

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NOTES

1. Input pulse parameters
   PWA = 500 ns ±10%
   PRP = 1,000 ns +10%
   t_{DELAY} = 250 ±50 ns
   $t_{TLHI} = t_{THLI}$ = 6 +2 ns
   PWB = 21 ±1 ns
2. Verify that all outputs have a high level ≥ 2 5 V and a low level ≤ 0 5 V before making the measurements specified in tables XV and XVI
3. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes YO1-1 and YO1-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25 4 mm)
4. All voltages are referenced to ground

* FIGURE 22 Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XV and XVI

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* TABLE XVII. Propagation delay time (t_{PLH}) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	CP to Q1 Propagation delay time (t_{PLH}), C_L = 50 pF	22	3,7,8,12,13,16,17,18, 21,23,24,26,27,32,33, 35,37	4	6,19, 36	5	27	ns	23,24
Q2	CP to Q2	24	3,7,8,12,13,16,17,18, 21,22,23,26,27,32,33, 35,37	25					
Q3	CP to Q3	3	7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37	5					
Q4	CP to Q4	23	3,7,8,12,13,16,17,18, 21,22,24,26,27,32,33, 35,37	28					
Q5	CP to Q5	8	3,7,12,13,16,17,18, 21,22,23,24,26,27,32, 33,37	11					
Q6	CP to Q6	7	3,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37	30					
Q7	CP to Q7	27	3,7,8,12,13,16,17,18, 21,22,23,24,26,32,33, 35,37	29					
Q8	CP to Q8	26	3,7,8,12,13,16,17,18, 21,22,23,24,27,32,33, 35,37	9					
Q9	CP to Q9	12	3,7,8,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	31					

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* TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q10	CP to Q10 Propagation delay time (t_{PLH}), C_L = 50 pF	32	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37	14	6,19, 36	5	27	ns	23,24
Q11	CP to Q11	33	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 35,37	34					
Q12	CP to Q12	13	3,7,8,12,16,17,18,21, 22,23,24,26,27,32,33, 35,37	15					
Q13	CP to Q13	18	3,7,8,12,13,16,17,21, 22,23,24,26,27,32,33, 35,37	39					
Q14	CP to Q14	37	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35	20					
Q15	CP to Q15	17	3,7,8,12,13,16,18,21, 22,23,24,26,27,32,33, 35,37	40					
Q16	CP to Q16	16	3,7,8,12,13,17,18,21, 22,23,24,26,27,32,33, 35,37	38					
Q17	CP to Q1	21	3,7,8,12,13,16,17,18, 22,23,24,26,27,32,33, 35,37	4	19,36				
Q18	CP to Q16	35	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,36,37	38	6,19				

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* TABLE XVII Propagation delay time (t_{PLH}) test - Continued

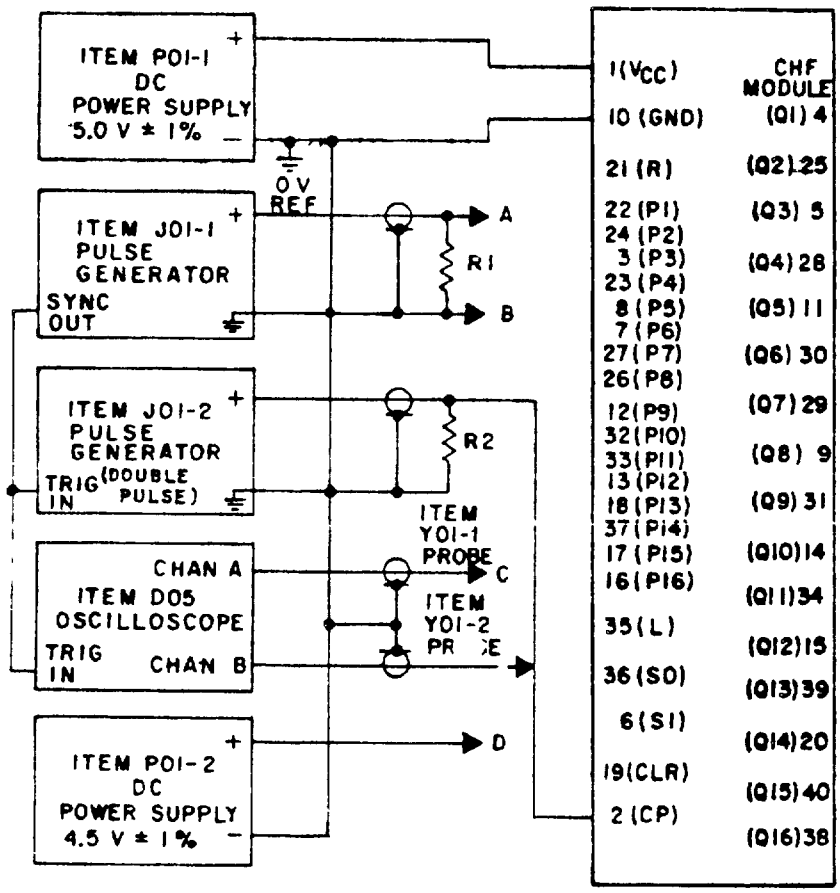
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q19	CP to Q1 Propagation delay time (t_{PLH}), C_L = 150 pF	22	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37	4	6,19, 36	5	32	ns	23,24
Q20	CP to Q2	24	3,7,8,12,13,16,17,18, 21,22,23,26,27,32,33, 35,37	25					
Q21	CP to Q3	3	7,8,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	5					
Q22	CP to Q4	23	3,7,8,12,13,16,17,18, 21,22,24,26,27,32,33, 35,37	28					
Q23	CP to Q5	8	3,7,1? 13,16,17,18,21, 22,23,_4,26,27,32,33, 35,3'	11					
Q24	CP to Q6	7	3,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37	30					
Q25	CP to Q7	27	3,7,8,12,13,16,17,18, 21,22,23,24,26,32,33, 35,37	29					
Q26	CP to Q8	26	3,7,8,12,13,16,17,18, 21,22,23,24,27,32,33, 35,37	9					
Q27	CP to Q9	12	3,7,8,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37	31					

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* TABLE XVII Propagation delay time (t_{PLH}) test - Continued

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q28	CP to Q10 Propagation delay time (t_{PLH}), C_L = 150 pF	32	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,33, 35,37	14	6,19, 36	5	32	ns	23,24
Q29	CP to Q11	33	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 35,37	34					
Q30	CP to Q12	13	3,7,8,12,16,17,18,21, 22,23,24,26,27,32,33, 35,37	15					
Q31	CP to Q13	18	3,7,8,12,13,16,17,21, 22,23,24,26,27,32,33, 35,37	39					
Q32	CP to Q14	37	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35	20					
Q33	CP to Q15	17	3,7,8,12,13,16,18,21, 22,23,24,26,27,32,33, 35,37	40					
Q34	CP to Q16	16	3,7,8,12,13,17,18,21, 22,23,24,26,27,32,33, 35,37	38					
Q35	CP to Q1	21	3,6,7,8,12,13,16,17, 18,22,23,24,26,27,32, 33,35,37	4	19,36				
Q36	CP to Q16	35	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,36,37	38	6,19				

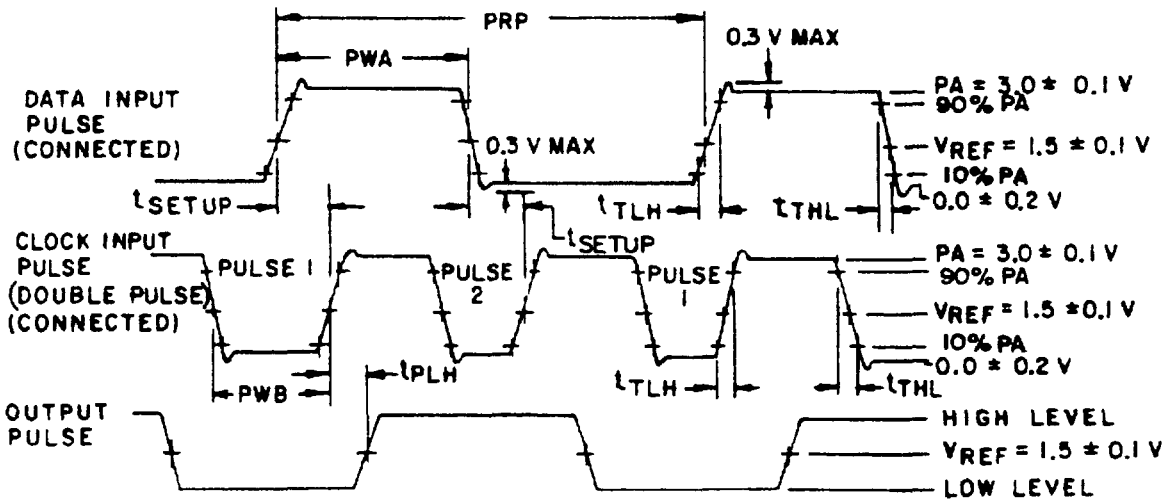
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NOTES
1 R1 = R2 = 51Ω ±5%, 1/2 W
2 Load each output with a load as specified on figure 17

* FIGURE 23 Propagation delay time (t_{PLH}) test circuit for table XVII

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NOTES
1. Input pulse parameters
   PRP = 500 ns ±10%
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns
   PWB = 100 ±10 ns
   $t_{SETUP} = 20 +2, -0$ ns
   PWA = 250 ±50 ns
2. Before making any measurements specified in table XVII, the setup time requirement must be verified
3. Physical length of electrical conductors between resistors R1 and R2 and input pin, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)
4. All voltages are referenced to ground.

* FIGURE 24 Propagation delay time (t_{PLH}) test circuit waveforms for table XVII

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* TABLE XVIII Propagation delay time (t_{PHL}) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	CP to Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	22	Open	4	3,7,8,12,13,16,17,18, 21,23,24,26,27,32,33, 35,37,6,19,36	5	31	ns	25,26
R2	CP to Q2	24		25	3,7,8,12,13,16,17,18, 21,22,23,26,27,32,33, 35,37,6,19,36				
R3	CP to Q3	3		5	7,8,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R4	CP to Q4	23		28	3,7,8,12,13,16,17,18, 21,22,24,26,27,32,33, 35,37,6,19,36				
R5	CP to Q5	8		11	3,7,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R6	CP to Q6	7		30	3,8,12,13,16,17,18,21, 22,23,24,26,27,32,35, 37,6,19,36				
R7	CP to Q7	27		29	3,7,8,12,13,16,17,18, 21,22,23,24,26,32,33, 35,37,6,19,36				
R8	CP to Q8	26		9	3,7,8,12,13,16,17,18, 21,22,23,24,27,32,33, 35,37,6,19,36				
R9	CP to Q9	12		31	3,7,8,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				

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* TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R10	CP to Q10 Propagation delay time (t_{PHL}), C_L = 50 pF	32	Open	14	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,33, 35,37,6,19,36	5	31	ns	25,26
R11	CP to Q11	33		34	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 35,37,6,19,36				
R12	CP to Q12	13		15	3,7,8,12,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R13	CP to Q13	18		39	3,7,8,12,13,16,17,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R14	CP to Q14	37		20	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,6,19,36				
R15	CP to Q15	17		40	3,7,8,12,13,16,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R16	CP to Q16	16		38	3,7,8,12,13,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R17	CP to Q1	21	6	4	3,7,8,12,13,17,18,22, 23,24,26,27,32,33,35, 37,16,19,36				
R18	CP to Q16	35	36	38	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,37,6,19				

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* TABLE XVIII Propagation delay time (t_{PHL}) test - Continued

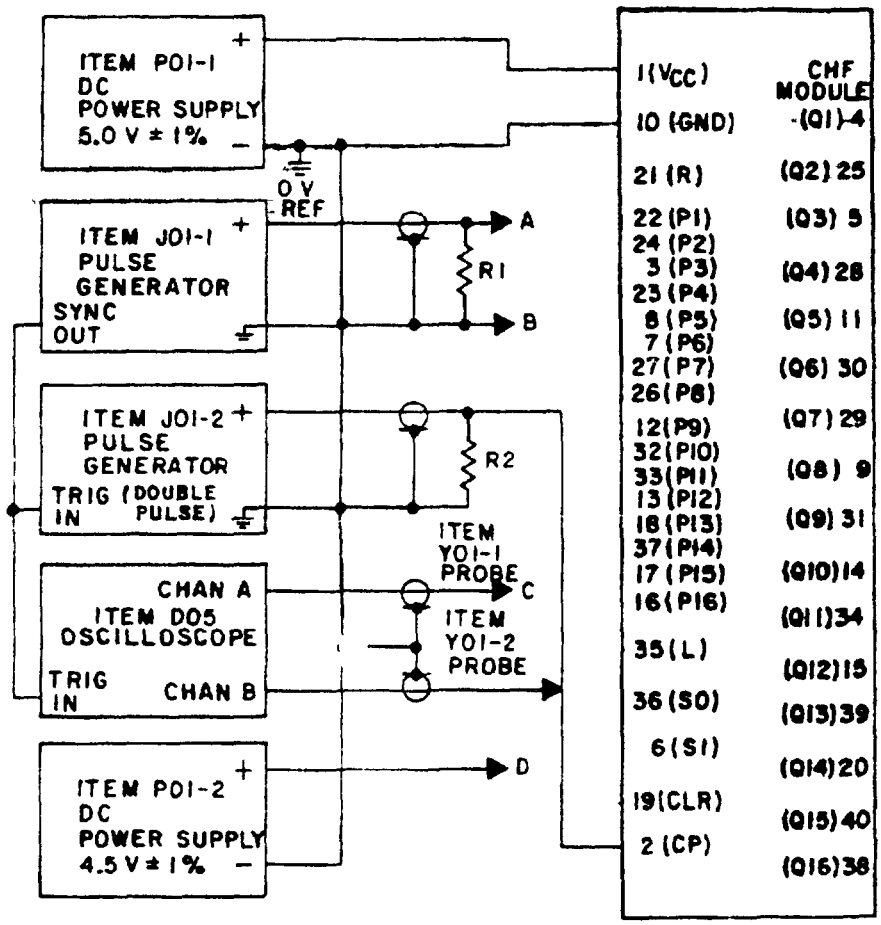
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R19	CP to Q1 Propagation delay time (t_{PHL}), C_L = 150 pF	22	Open	4	3,7,8,12,13,16,17,18, 21,23,24,26,27,32,33, 35,37,6,19,36	5	41	ns	25,26
R20	CP to Q2	24		25	3,7,8,12,13,16,17,18, 21,22,23,26,27,32,33, 35,37,6,19,36				
R21	CP to Q3	3		5	7,8,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R22	CP to Q4	23		28	3,7,8,12,13,16,17,18, 21,22,24,26,27,32,33, 35,37,6,19,36				
R23	CP to Q5	8		11	3,7,12,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R24	CP to Q6	7		30	3,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,37,6,19,36				
R25	CP to Q7	27		29	3,7,8,12,13,16,17,18, 21,22,23,24,26,32,33, 35,37,6,19,36				
R26	CP to Q8	26		9	3,7,8,12,13,16,17,18, 21,22,23,24,27,32,33, 35,37,6,19,36				
R27	CP to Q9	12		31	3,7,8,13,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				

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* TABLE XVIII Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R28	CP to Q10 Propagation delay time (t_{PHL}), C_L = 150 pF	32	Open	14	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,33, 35,37,6,19,36	5	41	ns	25,26
R29	CP to Q11	33		34	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,33, 35,37,6,19,36				
R30	CP to Q12	13		15	3,7,8,12,16,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R31	CP to Q13	18		39	3,7,8,12,13,16,17,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R32	CP to Q14	37		20	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,35,6,19,36				
R33	CP to Q15	17		40	3,7,8,12,13,16,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R34	CP to Q16	16		38	3,7,8,12,13,17,18,21, 22,23,24,26,27,32,33, 35,37,6,19,36				
R35	CP to Q1	21	6	4	3,7,8,12,13,16,17,18, 22,23,24,26,27,32,33, 35,37,19,36				
R36	CP to Q16	35	36	38	3,7,8,12,13,16,17,18, 21,22,23,24,26,27,32, 33,37,6,19				

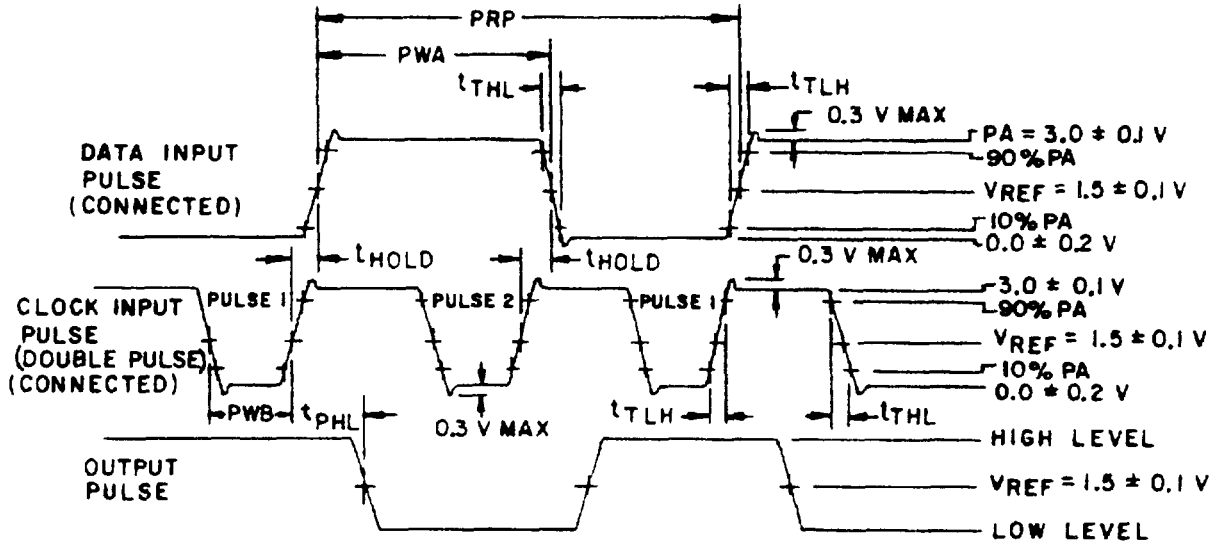
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. Load each output with a load as specified on figure 17

* FIGURE 25 Propagation delay time (t_{PHL}) test circuit for table XVIII

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NOTES
1 Input pulse parameters
PRP = 500 ns ±10%
PWB = 100 ±10 ns
$t_{TLH} = t_{THL} = 6 \pm 2$ ns
$t_{HOLD} = 5\ +2, -0$ ns
PWA = 250 ±50 ns
2 Before making any measurements specified in table XVIII, the hold time requirement must be verified
3 Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes YO1-1 and YO1-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)
4 All voltages are referenced to ground

*FIGURE 26 Propagation delay time (t_{PHL}) test circuit waveforms for table XVIII

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* TABLE XIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	CLR to Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	6, 36	Open	4	5	35	ns	27,28
S2	CLR to Q2			25				
S3	CLR to Q3			5				
S4	CLR to Q4			28				
S5	CLR to Q5			11				
S6	CLR to Q6			30				
S7	CLR to Q7			29				
S8	CLR to Q8			9				
S9	CLR to Q9			31				
S10	CLR to Q10			14				
S11	CLR to Q11			34				
S12	CLR to Q12			15				
S13	CLR to Q13			39				
S14	CLR to Q14			20				
S15	CLR to Q15			40				
S16	CLR to Q16			38				
S17	CLR to Q1	36	6	4				

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

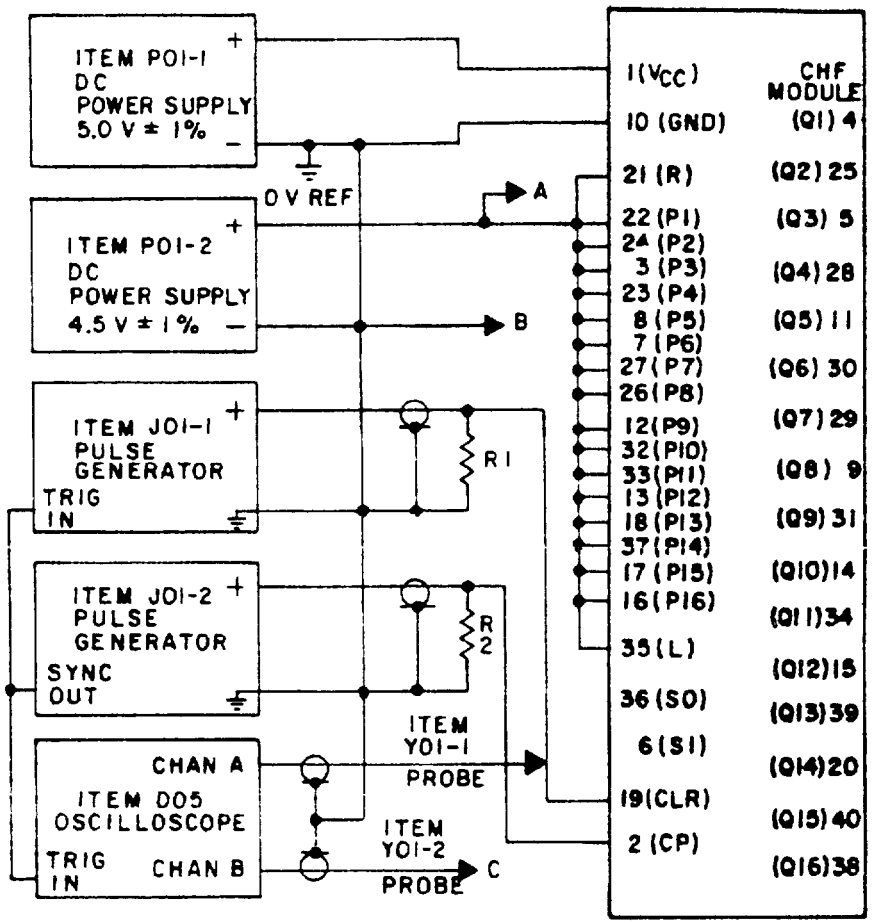
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S18	CLR to Q16 Propagation delay time (t_{PHL}), C_L = 50 pF	6	36	38	5	35	ns	27,28
S19	CLR to Q1 Propagation delay time (t_{PHL}), C_L = 150 pF	6, 36	Open	4		45		
S20	CLR to Q2			25				
S21	CLR to Q3			5				
S22	CLR to Q4			28				
S23	CLR to Q5			11				
S24	CLR to Q6			30				
S25	CLR to Q7			29				
S26	CLR to Q8			9				
S27	CLR to Q9			31				
S28	CLR to Q10			14				
S29	CLR to Q11			34				
S30	CLR to Q12			15				
S31	CLR to Q13			39				
S32	CLR to Q14			20				
S33	CLR to Q15			40				

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S34	CLR to Q16 Propagation delay time (t_{PHL}), C_L = 150 pF	6, 36	Open	38	5	45	ns	27,28
S35	CLR to Q1 Propagation delay time (t_{PHL}), C_L = 150 pF	36	6	4	5	45	ns	27,28
S36	CLR to Q16 Propagation delay time (t_{PHL}), C_L = 150 pF	6	36	38	5	45	ns	27,28

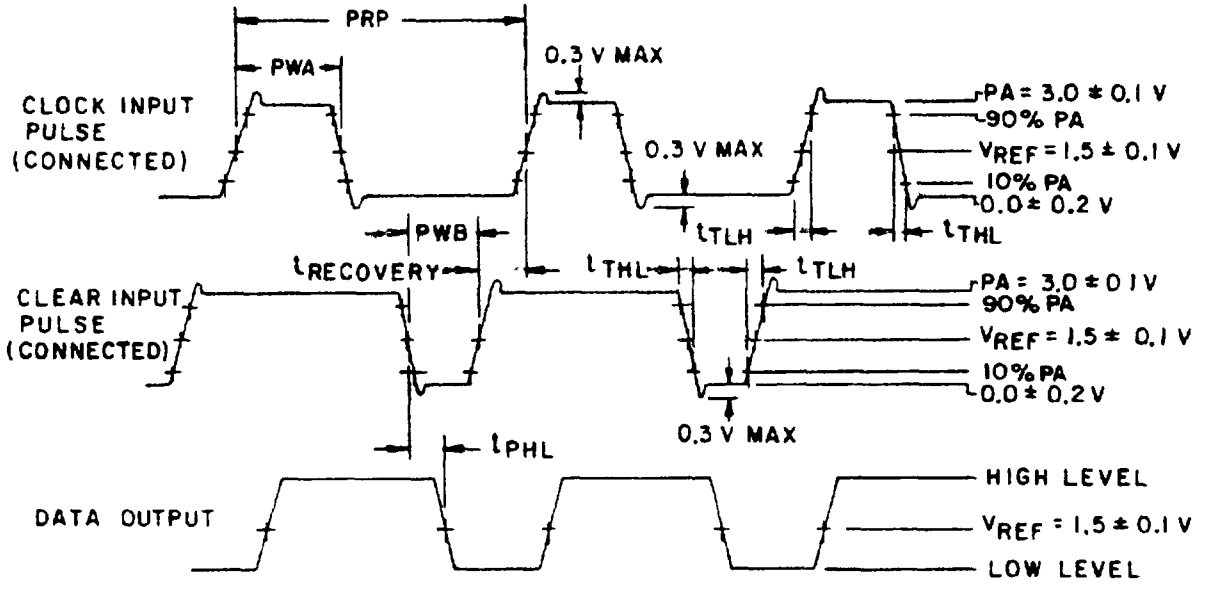
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NOTES
1 R1 = R2 = 51Ω ±5%, 1/2 W
2 Load each output with a load as specified on figure 17

*FIGURE 27 Propagation delay time (t_{PHL}) test circuit for table XIX

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NOTES
1 Clock input pulse parameters
PWA = 100 ±10 ns
PRP = 500 ±50 ns
$t_{TLH} = t_{THL} = 6 \pm 2$ ns
PWB = 26 ±1 ns (minimum clear pulse width)
$t_{RECOVERY} = 30 +2, -0$ ns (clear recovery time)
2 Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)
3 All voltages are referenced to ground.

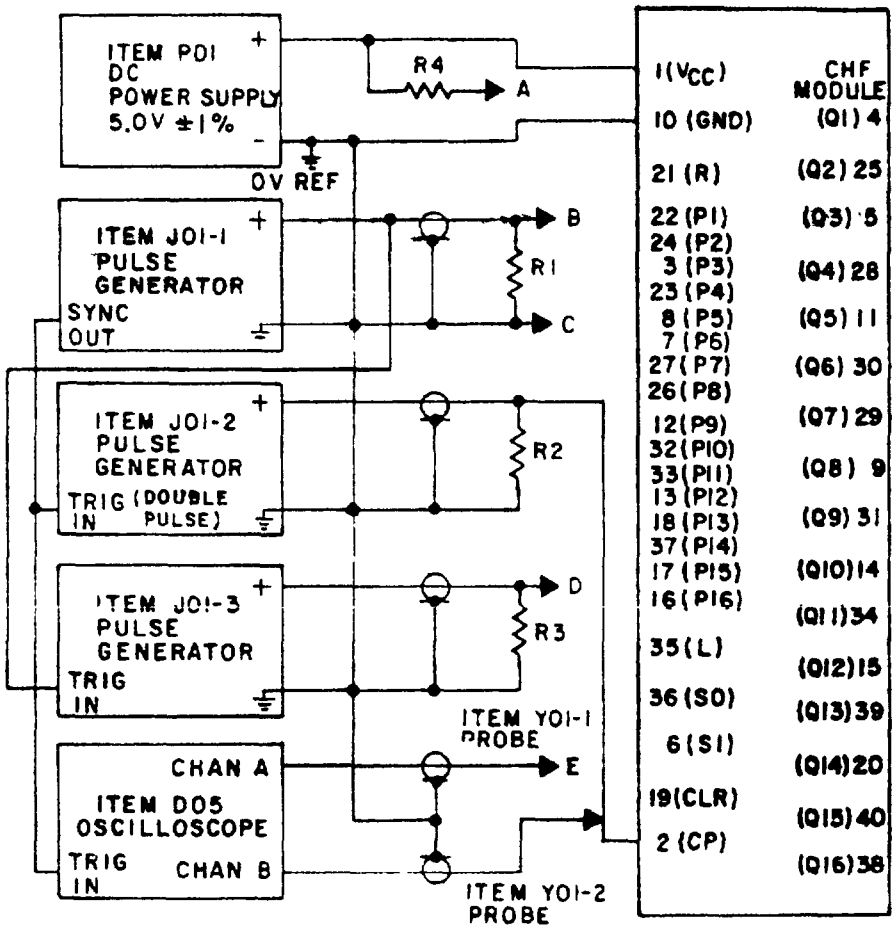
*FIGURE 28 Propagation delay time (t_{PHL}) test circuit waveforms for table XIX

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TABLE XX. Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check test

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	SO Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q1 output voltage, high level)	19	21	6,22	36	4	2.5	4.5	V	29,30
T2	S1 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q16 output voltage, low level)	16,19, 21	35	36	6	38	0.0	0.5		29,31
T3	SO,S1 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q8 output voltage, high level)	19	26	21,35	6,36	9	2.5	4.5		29,30
T4	SO,S1 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check (measuring Q9 output voltage, low level)	19,21, 35	12	Open	6,36	31	0.0	0.5		29,31

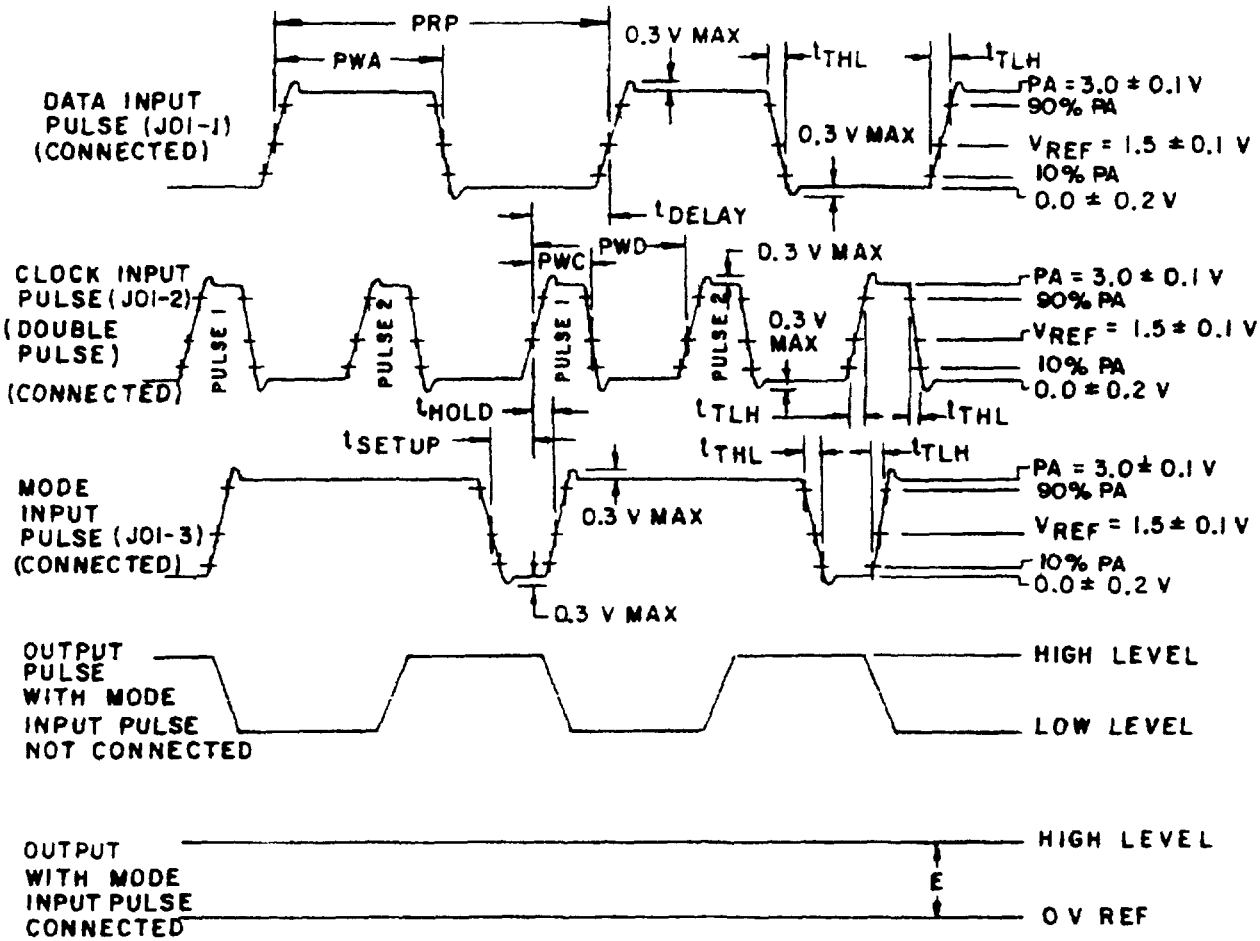
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NOTES
1 R1 - R3 = 51Ω ±5%, 1/2 W
2 R4 = 1 kΩ ±5%, 1/2 W
3 Load each output with a load as specified on figure 17

FIGURE 29 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check test circuit for table XX

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NOTES

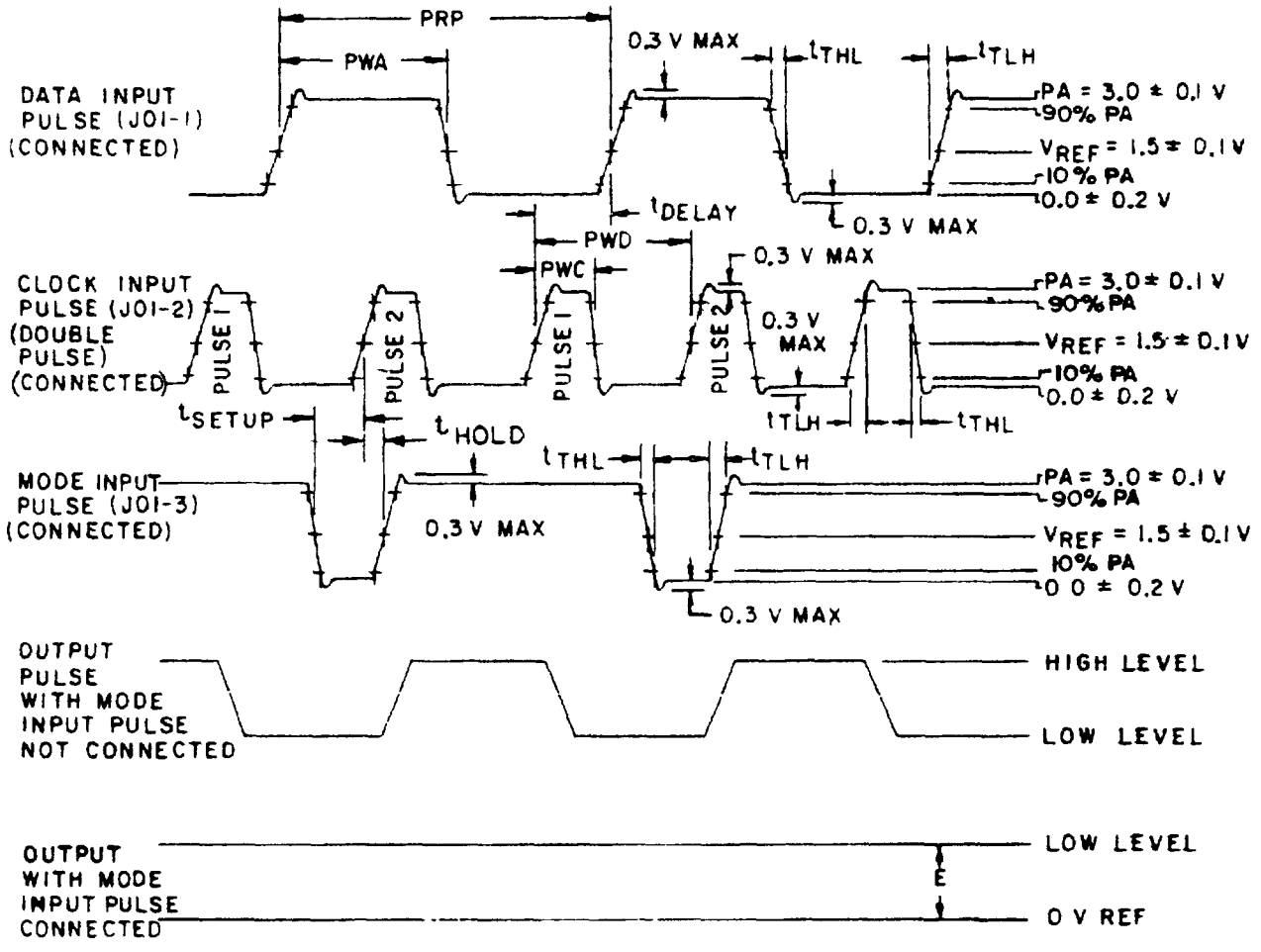
1 Input pulse parameters
- PWA = 250 ±10 ns
- PRP = 500 ±25 ns
- $t_{THL} = t_{TLH} = 6 \pm 2$ ns
- PWC = 100 ±10 ns
- PWD = 250 ±10 ns.
- $t_{HOLD} = 0$ +2, -0 ns
- $t_{SETUP} = 30$ +2, -0 ns
- $t_{DELAY} = 75$ ±10 ns

2 Physical length of electrical conductors between resistors R1, R2, and R3 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)

3 All voltages are referenced to ground

* FIGURE 30 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check test circuit waveform for table XX

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NOTES

1 Data input pulse parameters
- PWA = 250 ±10 ns.
- PRP = 500 ±25 ns
- $t_{THL} = t_{TLH} = 6 \pm 2$ ns
- PWC = 100 ±10 ns
- PWD = 250 ±10 ns
- $t_{HOLD} = 0$ +2, -0 ns
- $t_{SETUP} = 30$ +2, -0 ns
- $t_{DELAY} = 75 \pm 10$ ns

2 Physical length of electrical conductors between resistors R1, R2, and R3 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)

3 All voltages are referenced to ground

* FIGURE 31 Mode input setup (t_{SETUP}) and hold (t_{HOLD}) time check test circuit waveforms for table XX

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* TABLE XXI Maximum operating frequency check test.

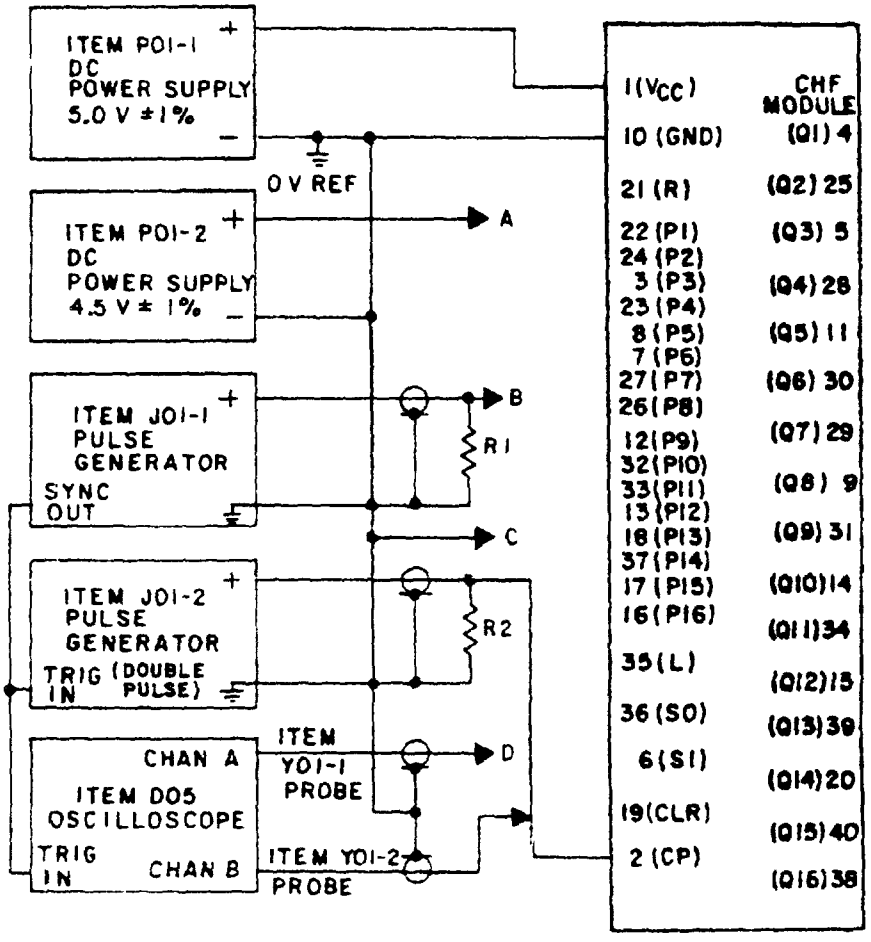
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Q1 Maximum operating frequency check (t_{PW})	6,19,21, 35,36	22	Open	4	112	138	ns	32,33
U2	Q2		24		25				
U3	Q3		3		5				
U4	Q4		23		28				
U5	Q5		8		11				
U6	Q6		7		30				
U7	Q7		27		29				
U8	Q8		26		9				
U9	Q9		12		31				
U10	Q10		32		14				
U11	Q11		33		34				
U12	Q12		13		15				
U13	Q13		18		39				
U14	Q14		37		20				
U15	Q15		17		40				
U16	Q16		16		38				

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* TABLE XXI. Maximum operating frequency check test - Continued

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U17	Q16 Maximum operating frequency check (t_{pW})	19,35,36	21	6	38	112	138	ns	32,33
U18	Q1 Maximum operating frequency check (t_{pW})	6,19,21	35	36	4	112	138	ns	32,33

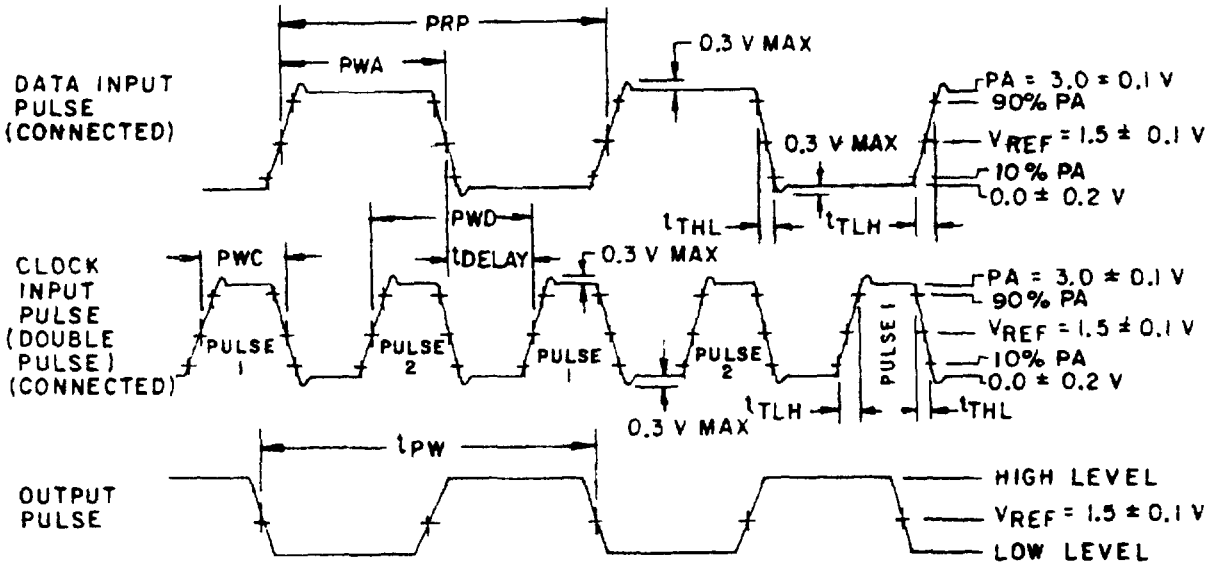
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. Load each output with a load as specified on figure 17

FIGURE 32. Maximum operating frequency check test circuit for table XXI

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NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL} = 6 \pm 2$ ns
   PWA = 62 ±2 ns
   PRP = 125 ±4 ns
   PWC = 31 ±1 ns
   PWD = 62 ±2 ns (minimum clock period)
   $t_{DELAY} = 31 \pm 1$ ns
2. All voltages are referenced to ground

FIGURE 33 Maximum operating frequency check test circuit waveforms for table XXI

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	Q1 Output crosstalk, low level	21,35, 36,6, 19	3,7,8,12, 13,16,17, 18,23,24, 26,27,32, 33,37	22	25	4	0	650	mV	34,35
V2	Q2		3,7,8,12, 13,16,17, 18,22,23, 26,27,32, 33,37	24	4	25				
V3	Q3		7,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	1	25	5				
V4	Q4		3,7,8,12, 13,16,17, 18,22,24, 26,27,32, 33,37	23	25	28				
V5	Q5		3,7,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	8	30	11				
V6	Q6		3,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	7	11	30				
V7	Q7		3,7,8,12, 13,16,17, 18,22,23, 24,26,32, 33,37	27	30	29				

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V8	Q8 Output crosstalk, low level	21,35, 36,6, 19	3,7,8,12, 13,16,17, 18,22,23, 24,27,32, 33,37	26	30	9	0	650	mV	34,35
V9	Q9		3,7,8,13, 16,17,18, 22,23,24, 26,27,32, 33,37	12	14	31				
V10	Q10		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 33,37	32	31	14				
V11	Q11		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,37	33	14	34				
V12	Q12		3,7,8,12, 16,17,18, 22,23,24, 26,27,32, 33,37	13	14	15				
V13	Q13		3,7,8,12, 13,16,17, 22,23,24, 26,27,32, 33,37	18	20	39				
V14	Q14		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33	37	39	20				

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TABLE XXII Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V15	Q15 Output crosstalk, low level	21,35, 36,6, 19	3,7,8,12, 13,16,18, 22,23,24, 26,27,32, 33,37	17	20	40	0	650	mV	34,35
V16	Q16 Output crosstalk, low level	21,35, 36,6, 19	3,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,37	16	20	38		650		34,35
V17	Q1 Output crosstalk, low level, duration	21,35, 36,6, 19	3,7,8,12, 13,16,17, 18,23,24, 26,27,32, 33,37	22	2	4		400		34,36
V18	Q2		3,7,8,12, 13,16,17, 18,22,23, 26,27,32, 33,37	24		25				
V19	Q3		7,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	3		5				
V20	Q4		3,7,8,12, 13,16,17, 18,22,24, 26,27,32, 33,37	23		28				
V21	Q5		3,7,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	8		11				

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V22	Q6 Output crosstalk, low level, duration	21,35, 36,6, 19	3,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	7	2	30	0	400	mV	34,36
V23	Q7		3,7,8,12, 13,16,17, 18,22,23, 24,26,32, 33,37	27		29				
V24	Q8		3,7,8,12, 13,16,17, 18,22,23, 24,27,32, 33,37	26		9				
V25	Q9		3,7,8,13 16,17,18, 22,23,2', 26,27,32, 33,37	12		31				
V26	Q10		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 33,37	32		14				
V27	Q11		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,37	33		34				
V28	Q12		3,7,8,12, 16,17,18, 22,23,24, 26,27,32, 33,37	13		15				

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V29	Q13 Output crosstalk, low level, duration	21,35, 36,6, 19	3,7,8,12, 13,16,17, 22,23,24, 26,27,32, 33,37	18	2	39	0	400	mV	34,36
V30	Q14		3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33	37		20				
V31	Q15		3,7,8,12, 13,16,18, 22,23,24, 26,27,32, 33,37	17		40				
V32	Q16		3,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,37	16		38				
W1	Q1 Output crosstalk, high level	21,22, 35,36, 6,19	3,7,8,12, 13,16,17, 18,23,24, 26,27,32, 33,37	Open	25	4				34,35
W2	Q2 Output crosstalk, high level	21,24, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,23, 26,27,32, 33,37	Open	4	25				34,35
W3	Q3 Output crosstalk, high level	21,3, 35,36, 6,19	7,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37	Open	25	5				34,35

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued

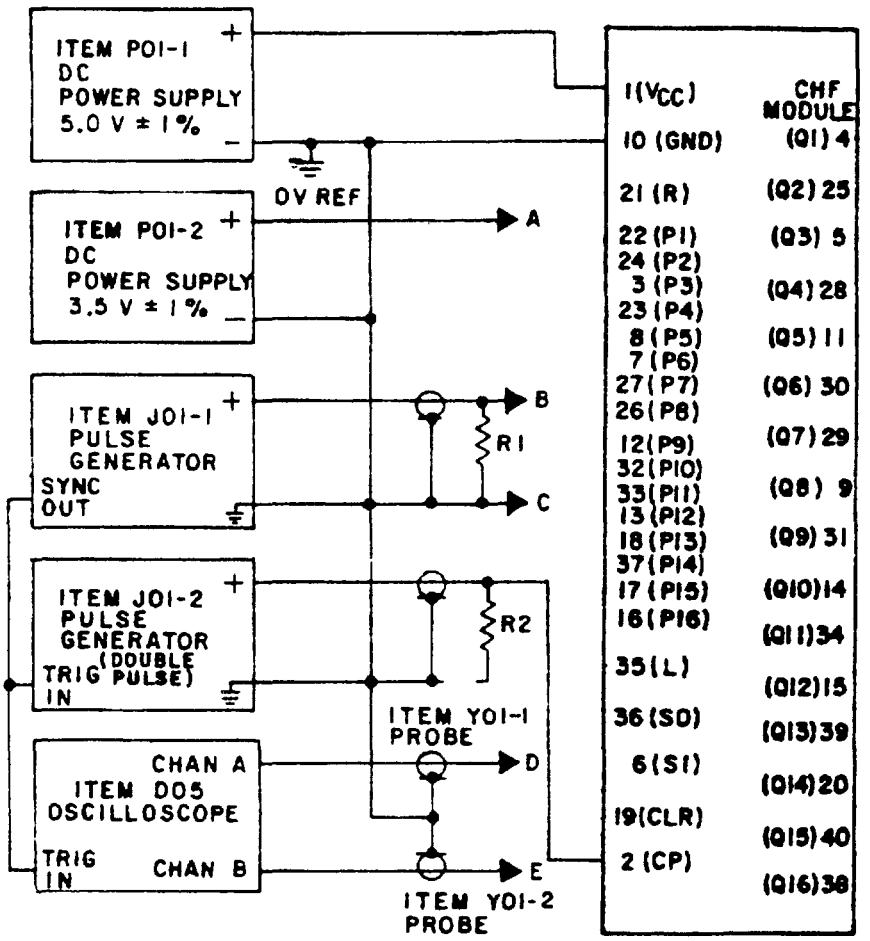
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W4	Q4 Output crosstalk, high level	21,23, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,24, 26,27,32, 33,37	Open	25	28	0	400	mV	34,35
W5	Q5	21,8, 35,36, 6,19	3,7,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37		30	11				
W6	Q6	21,7, 35,36, 6,19	3,8,12,13, 16,17,18, 22,23,24, 26,27,32, 33,37		11	30				
W7	Q7	21,27, 35,36, 6,19	3,7,8,1; 13,16,17, 18,22,23, 24,26,32, 33,37		30	29				
W8	Q8	21,26, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,23, 24,27,32, 33,37		30	9				
W9	Q9	21,12, 35,36, 6,19	3,7,8,13, 16,17,18, 22,23,24, 26,27,32, 33,37		14	31				
W10	Q10	21,32, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 33,37		31	14				

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TABLE XXII. Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W11	Q11 Output crosstalk, high level	21,33, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,37	Open	14	34	0	400	mV	34,35
W12	Q12	21,13, 35,36, 6,19	3,7,8,12, 16,17,18, 22,23,24, 26,27,32, 33,37		14	15				
W13	Q13	21,18, 35,36, 6,19	3,7,8,12, 13,16,17, 22,23,24, 26,27,32, 33,37		20	39				
W14	Q14	21,37, 35,36, 6,19	3,7,8,12, 13,16,17, 18,22,23, 24,26,27, 32,33		39	20				
W15	Q15	21,17, 35,36, 6,19	3,7,8,12, 13,16,18, 22,23,24, 26,27,32, 33,37		20	40				
W16	Q16	21,16, 35,36, 6,19	3,7,8,12, 13,17,18, 22,23,24, 26,27,32, 33,37		20	38				

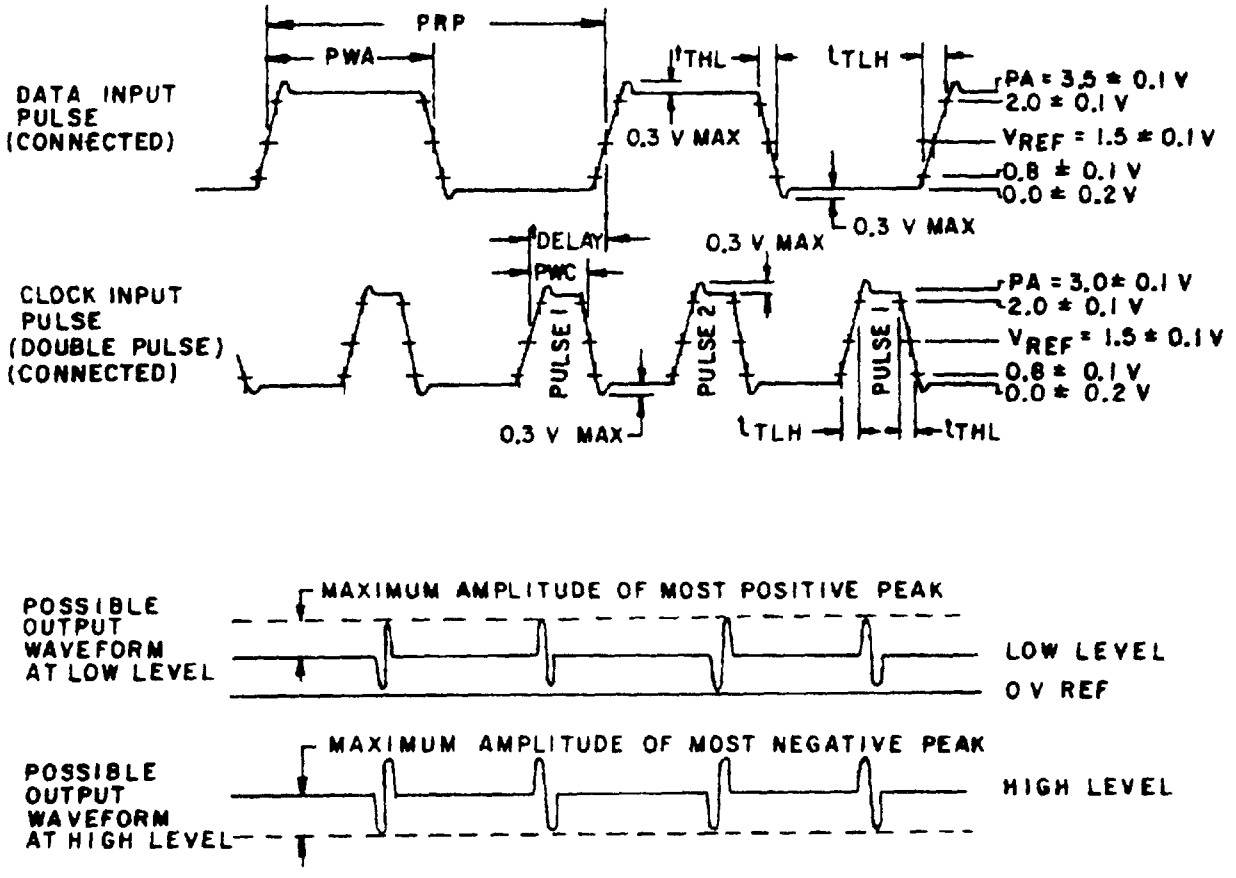
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NOTES
1 R1 = R2 = 51Ω ±5%, 1/2 W
2 Load all outputs with a load as specified on figure 17 with C_L = 50 pF

FIGURE 34 Output crosstalk (low-level and high-level) and output crosstalk (low-level) duration test circuit for table XXII

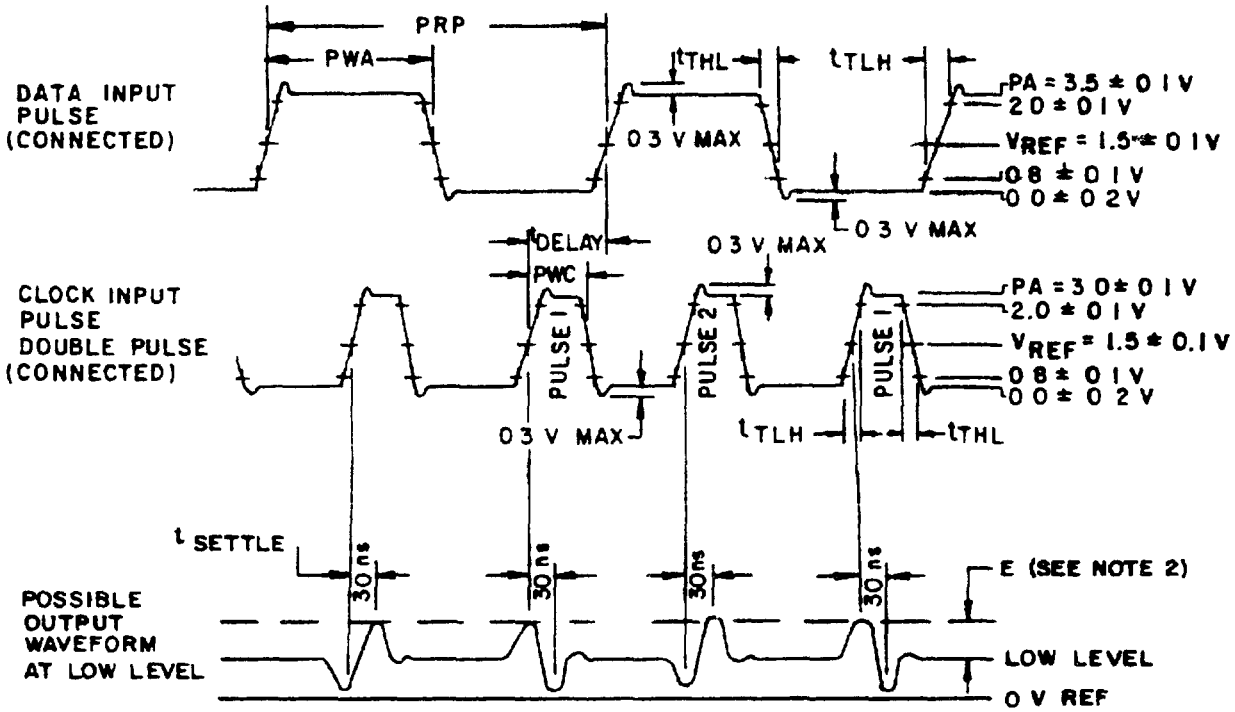
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NOTES
1. Input pulse parameters
   PWA = 100 ±10 ns
   PRP = 200 ±20 ns
   PWC = 25 ±5 ns
   $t_{TLH} = t_{THL}$ = 3 5 ±1 5 ns (verified at each input pin)
   t_{DELAY} = 50 ±5 ns
2. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes YO1-1 and YO1-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25 4 mm)
3. All voltages are referenced to ground

FIGURE 35 Output crosstalk (low-level and high-level) test circuit waveforms for table XXII

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NOTES

1. Input pulse parameters
   PWA = 100 ±10 ns
   PRP = 200 ±20 ns.
   PWC = 25 ±5 ns
   $t_{TLH} = t_{THL}$ = 3 5 ±1 5 ns (verified at each input pin)
   t_{DELAY} = 50 ±5 ns.
2. E is the maximum amplitude of the most positive peak measured after a settle time of 30 ns has elapsed after any positive transition of the clock pulse and before the next positive transition of the clock pulse.
3. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 00 inch (25 4 mm)
4. All voltages are referenced to ground

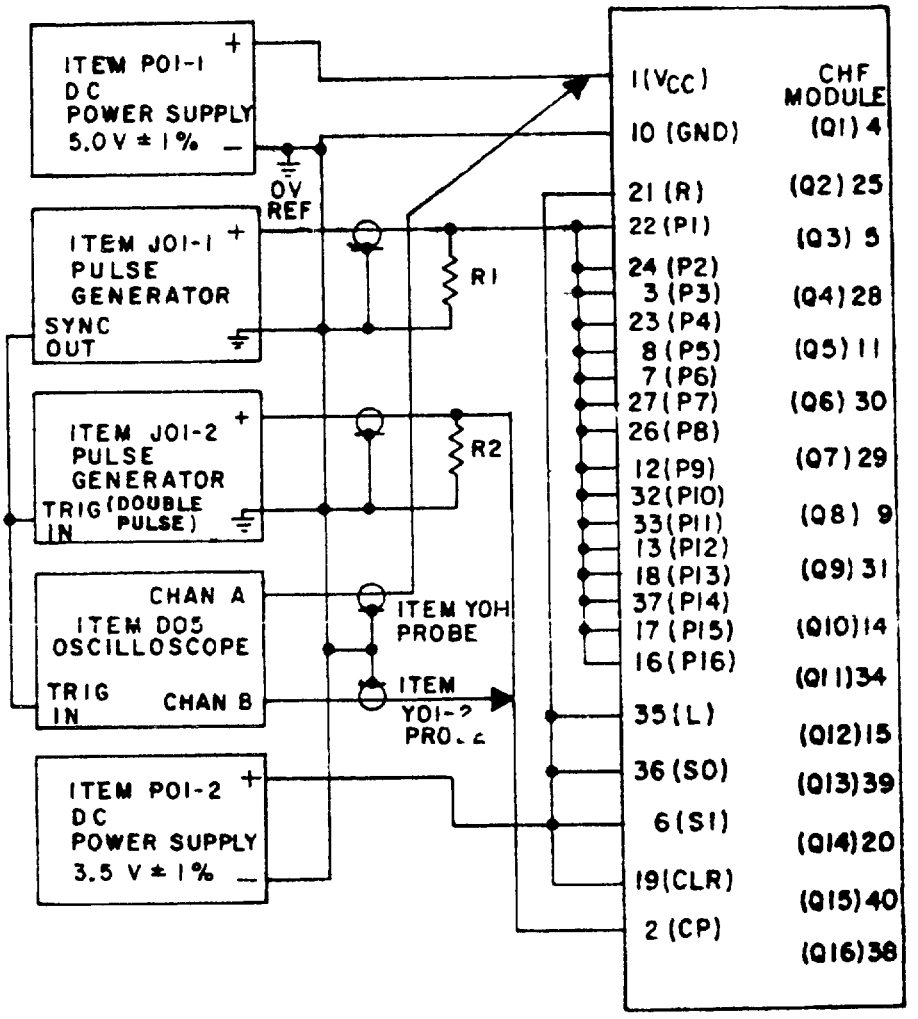
FIGURE 36. Output crosstalk (low-level) duration test circuit waveforms for table XXII.

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TABLE XXIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	37,38

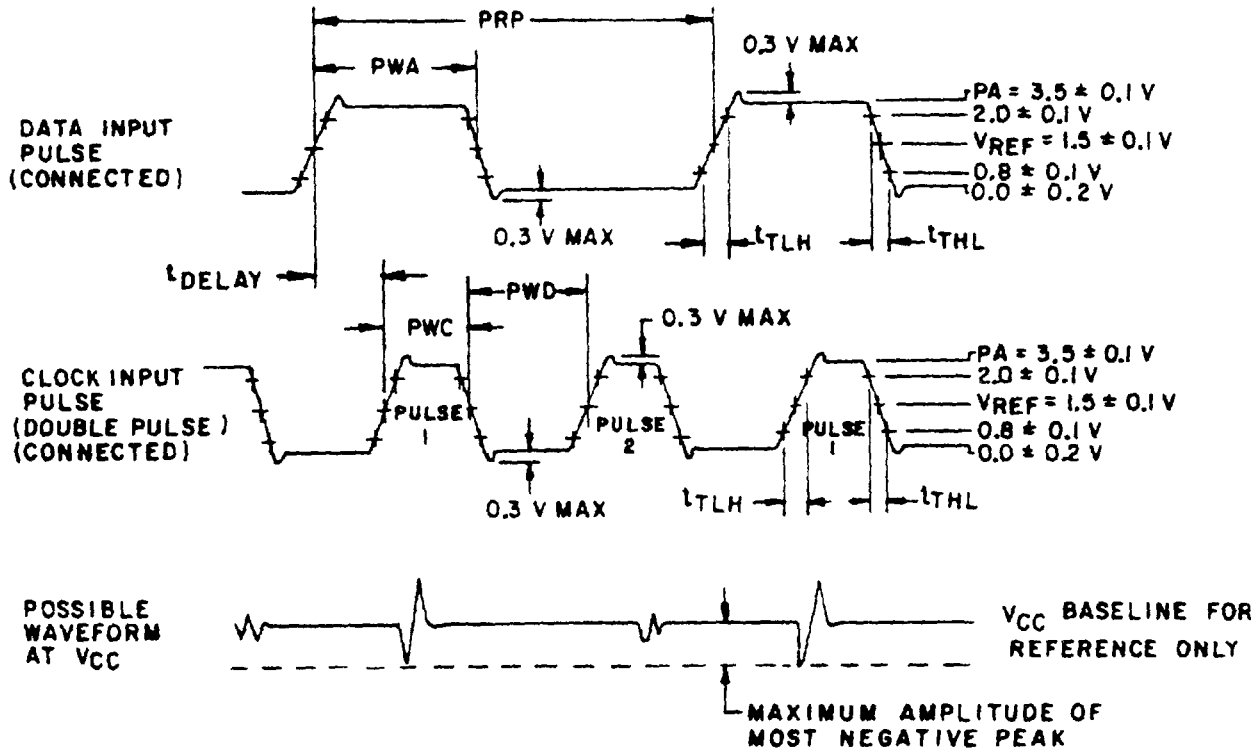
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NOTE R1 = R2 = 51Ω ±5%, 1/2 W

FIGURE 37 Power supply transient amplitude test circuit for table XXIII

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NOTES
1 Input pulse parameters
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)
PWA = 500 ±50 ns
PRP = 1,000 ±100 ns
PWC = 250 ±25 ns
PWD = 250 ±25 ns
t_{DELAY} = 250 ±25 ns
2 All voltages are referenced to ground

*FIGURE 38 Power supply transient amplitude test circuit waveforms for table XXIII.

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TABLE XXIV. Power supply filter capacitance test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply filter capacitance	Open	1	1.6	3.0	µF	39

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TABLE XXV. Input capacitance test.

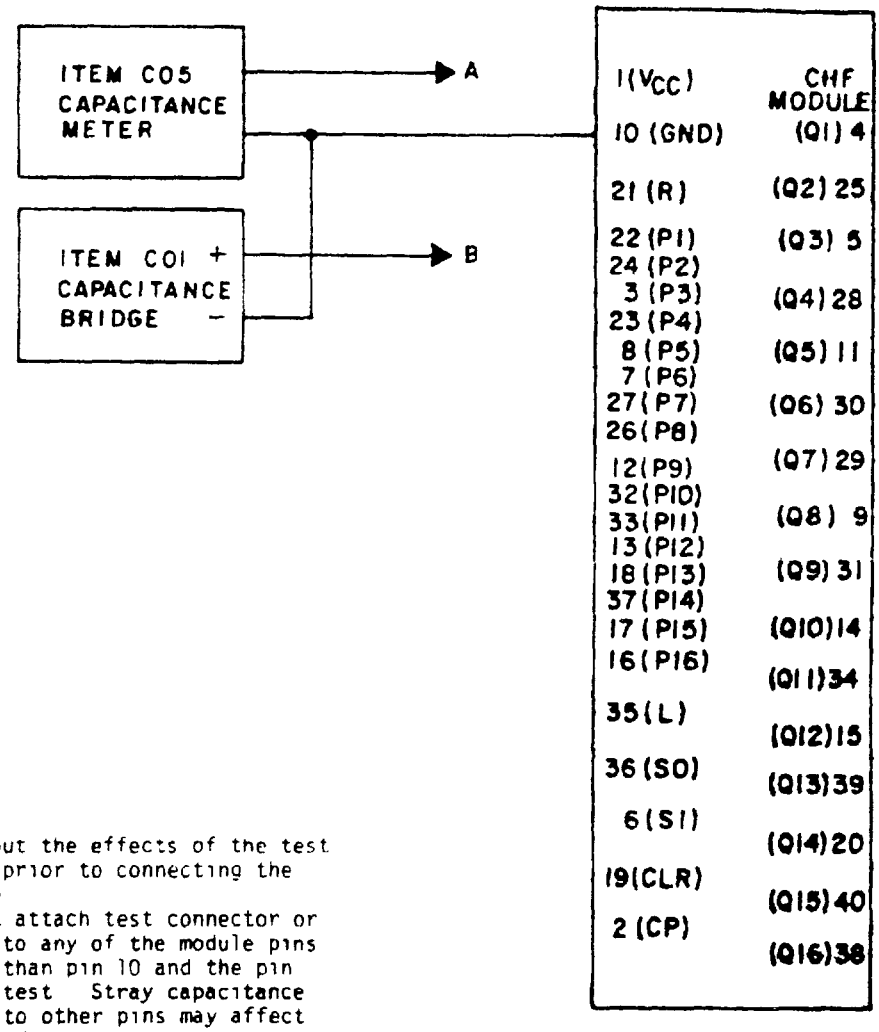
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z1	R Input capacitance	21	Open	2	20	pF	39
Z2	P1	22					
Z3	P2	24					
Z4	P3	3					
Z5	P4	23					
Z6	P5	8					
Z7	P6	7					
Z8	P7	27					
Z9	P8	26					
Z10	P9	12					
Z11	P10	32					
Z12	P11	33					
Z13	P12	13					
Z14	P13	18					
Z15	P14	37					
Z16	P15	17					
Z17	P16	16					
Z18	L	35					

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TABLE XXV Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z19	S0 Input capacitance	36	Open	2	70	pF	39
Z20	S1	6	↓	↓	↓	↓	↓
Z21	CLR	19					
Z22	CP	2					

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NOTES
1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3. Capacitance bridge (C01) setup
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum

FIGURE 39. Power supply filter and input capacitance test circuit for tables XXIV and XXV

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TABLE XXVI. Noise sensitivity test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	Noise sensitivity (CLR)	19	4	2	19	6,36	450	550	ns	40,41
AA2			25							
AA3			5							
AA4			28							
AA5			11							
AA6			30							
AA7			29							
AA8			9							
AA9			31							
AA10			14							
AA11			34							
AA12			15							
AA13			39							
AA14			20							
AA15			40							
AA16			38							

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TABLE XXVI Noise sensitivity test - Continued

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA17	Noise sensitivity (SO)	36	4	2	36	6,19	450	550	ns	40,41
AA18			25							
AA19			5							
AA20			28							
AA21			11							
AA22			30							
AA23			29							
AA24			9							
AA25			31							
AA26			14							
AA27			34							
AA28			15							
AA29			39							
AA30			20							
AA31			40							
AA32			38							

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TABLE XXVI. Noise sensitivity test - Continued.

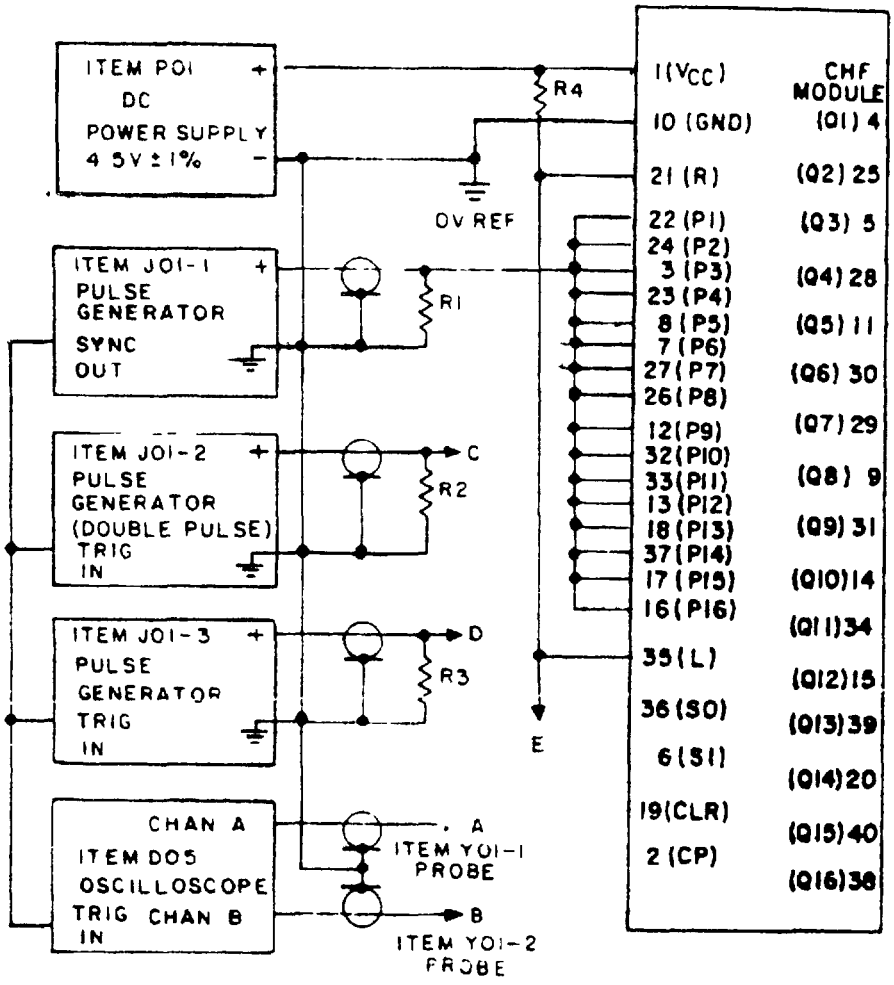
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA33	Noise sensitivity (S1)	6	4	2	6	19,36	450	550	ns	40,41
AA34			25							
AA35			5							
AA36			28							
AA37			11							
AA38			30							
AA39			29							
AA40			9							
AA41			31							
AA42			14							
AA43			34							
AA44			15							
AA45			39							
AA46			20							
AA47			40							
AA48			38							

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TABLE XXVI. Noise sensitivity test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA49	Noise sensitivity (CP)	2	4	Open	2	6,19,36	0.0	1.5	V	40,42
AA50			25							
AA51			5							
AA52			28							
AA53			11							
AA54			30							
AA55			29							
AA56			9							
AA57			31							
AA58			14							
AA59			34							
AA60			15							
AA61			39							
AA62			20							
AA63			40							
AA64			38							

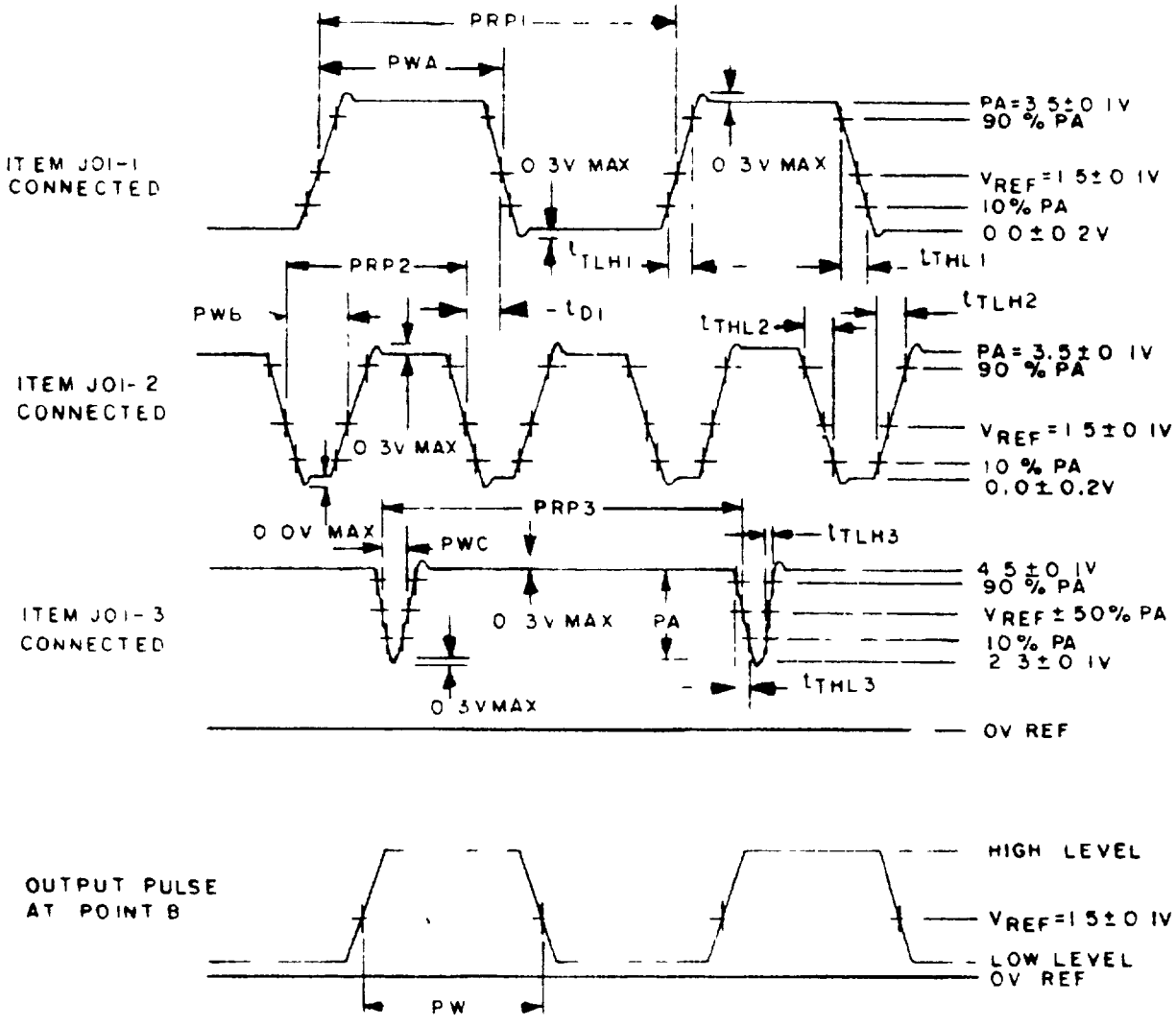
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NOTES
1. R1-R3 = 51.1Ω ±1%, 1/4 W.
2. R4 = 1 kΩ ±1%, 1/2 W.
3. Load each output with a load as specified on figure 17 with the exception that C_L = 50 ±5 pF (including probe and parasitic capacitance).
4. Clear the register before making measurements for test conditions AA49-AA64.

FIGURE 40. Noise sensitivity test circuit for table XXVI.

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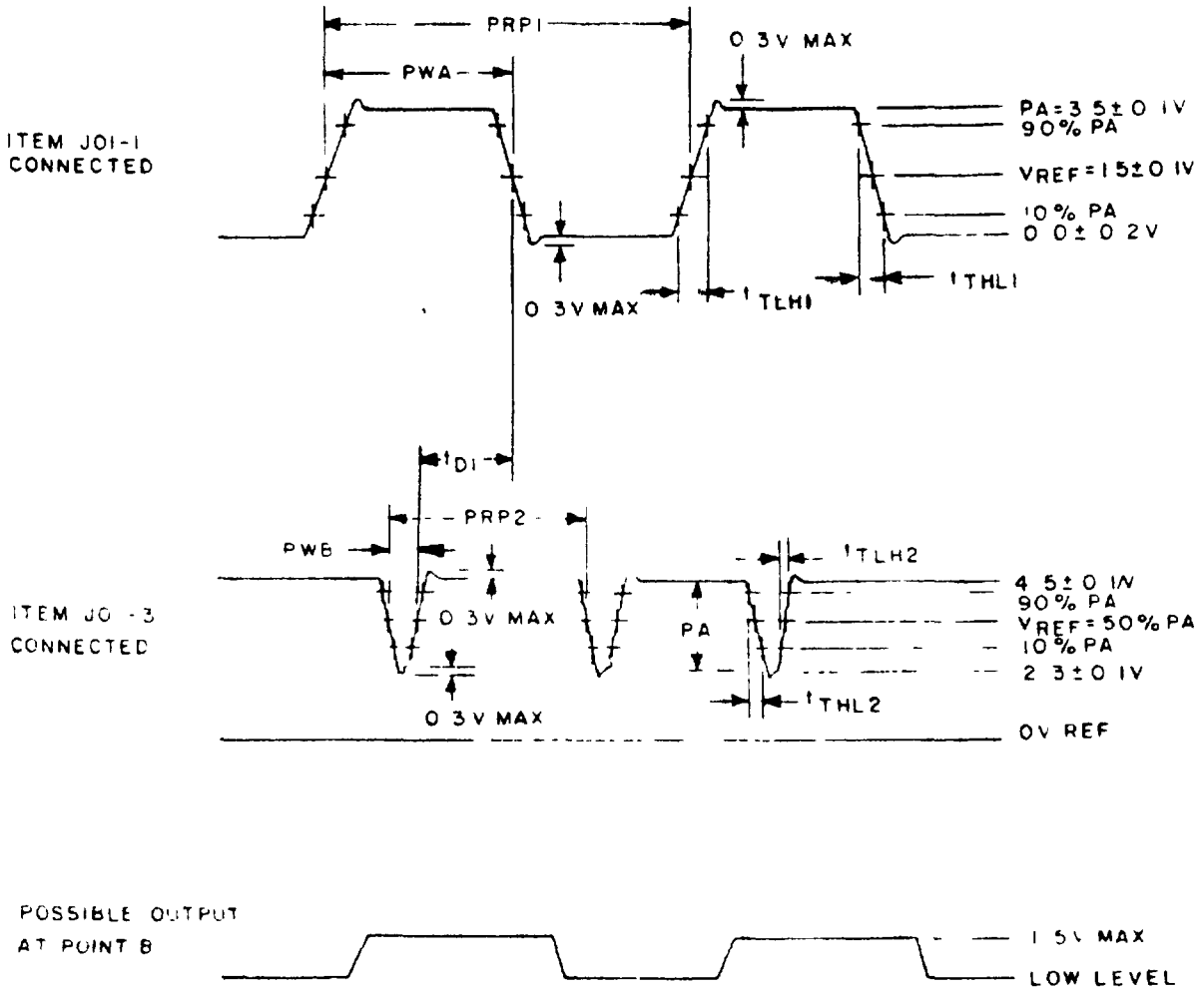


NOTES:

1. Item J01-1 pulse parameters
   PRP1 = 1000 ±100 ns.
   PWA = 500 ±50 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Item J01-2 pulse parameters.
   PRP2 = 500 ±50 ns.
   PWB = 200 ±20 ns.
   $t_{TLH2} = t_{THL2}$ = 10 ±2 ns.
   t_{D1} = 100 ±10 ns.
3. Item J01-3 pulse parameters
   PRP3 = 1000 ±100 ns.
   PWC = 20 ±5 ns.
   $t_{TLH3} = t_{THL3}$ = 6 ±2 ns.
4. Sweep the item J01-3 pulse through one cycle. The output pulse at point B shall meet the requirements of table XXVI (test conditions AA1-AA48).

FIGURE 41. Noise sensitivity test circuit waveforms for table XXVI.

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NOTES:
1. Item J01-1 pulse parameters
   PRP1 = 1000 ±100 ns.
   PWA = 500 ±50 ns.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   $t_{D1} = 100 \pm 10$ ns.
2. Item J01-3 pulse parameters
   PRP2 = 500 ±100 ns.
   PWB = 20 ±5 ns.
   $t_{TLH2} = t_{THL2} = 6 \pm 2$ ns.
3. The output pulse at point B shall meet the requirements of table XXVI (test conditions AA49-AA64).

FIGURE 42. Noise sensitivity test circuit waveforms for table XXVI.

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